

UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA

COS700 Research Report

An AI and Blockchain Framework for Multilingual Election Misinformation Monitoring in South Africa

Student numbers: u20790776, u26854563, u25753178

Supervisor(s):
Dr. S Rananga
Mr. N Rananga

Tumiso Mokautu Josh Cloete Tshiamo Monametsi

November 2026

Abstract

Election misinformation in South Africa spreads faster than any single organisation can verify it, often in languages automated tools do not support, and corrections are frequently dismissed as biased. Using design-science research, this study proposes a conceptual framework in which a multilingual AI model advises, independent fact-checking organisations decide jointly and a permissioned blockchain records their verdicts, and evaluates it through a working prototype. Because no labelled election corpus existed in Sepedi or isiZulu, statements were generated in English with Gemini 3.6 Flash and machine-translated, yielding 13,997 statements balanced across English, Sepedi and isiZulu. AfroXLMR-large-76L outperformed all other models tested, reaching a mean test F1-score of 0.899 with full fine-tuning and 0.882 with LoRA. The model labels every claim to help human fact-checkers prioritise, and their votes are recorded on a Hyperledger Fabric consortium ledger. Ledger writes are restricted in chaincode through X.509 certificate attributes, and two-thirds of member organisations must agree to confirm a claim as misinformation, with a simple-majority fallback for reports still open after 72 hours. Only hashes and metadata are stored on-chain, while claim documents stay off-chain in IPFS, aligning the design with the Protection of Personal Information Act (POPIA) at the architectural level. Hyperledger Caliper benchmarks with up to 20 organisations showed that report submission held its 25 TPS send rate at every consortium size, and identified full-ledger reads on CouchDB as the main performance bottleneck.

Keywords:

election misinformation, multilingual misinformation detection, low-resource languages, AfroXLMR, LoRA, permissioned blockchain, Hyperledger Fabric, fact-checking, POPIA, South Africa

Contents

1 Introduction **6**

2 Background **8**
2.1 Election Misinformation in South Africa 8
2.2 The Fact-Checking Ecosystem: Fragmentation and Trust . . . 9
2.3 AI for Multilingual Misinformation Detection 11
2.4 Blockchain as a Shared, Tamper-Evident Record 12
2.5 POPIA and the Immutability Paradox 13

3 Related Work **14**
3.1 Misinformation Detection in Low-Resource South African Languages 15
3.1.1 South African Detection Studies Built on Translated Data 15
3.1.2 Language Models for African-Language Classification . 15
3.1.3 LLM-Generated Training Data 16
3.1.4 Parameter-Efficient Fine-Tuning 16
3.2 Permissioned Blockchains for Shared Institutional Records . . 16
3.3 Shared Fact-Checking Infrastructure and Blockchain Approaches 17
3.4 Summary and Research Gap 18

4 Problem Statement and Requirements **19**
4.1 Problem Statement 19
4.2 Research Question 20
4.3 Sub-Research Questions 20
4.4 Research Objectives 21
4.5 Requirements 21
4.6 Scope and Limitations 22
4.7 Assumptions 23

5 Conceptual Framework **23**
5.1 Design Principles 24
5.2 Actors and Roles 25
5.3 Claim Lifecycle 25
5.4 Governance and Data Protection 27
5.5 From Framework to Prototype 27

6 Research Method and Instantiation **27**
6.1 Design-Science Approach 27
6.2 Synthetic Data Generation 29
6.2.1 Prompt Construction and Generation 29
6.2.2 Translation and Quality Control 30
6.2.3 Final Corpus 31
6.3 Misinformation Detection Model 31
6.3.1 Data Preparation 31
6.3.2 Models 32
6.3.3 Fine-Tuning 33
6.3.4 Evaluation and Model Selection 34
6.4 System Architecture 35
6.4.1 Blockchain Layer 35
6.4.2 Application Pipeline 37
6.5 Hyperledger Fabric Blockchain Framework 37
6.5.1 Permissioned Blockchain Justifications 37
6.5.2 Hyperledger Fabric Justification 38
6.6 Chaincode and On-Chain Data Model 38
6.6.1 Report Identity and Off-Chain Storage 38
6.6.2 Consensus and Finalisation Paths 39
6.7 Role-Based Access Control (RBAC) 41
6.8 Testing and Verification 42

7 Evaluation and Discussion **42**
7.1 AI-Enabled Misinformation Detection 42
7.1.1 Overall Performance 43
7.1.2 Full Fine-Tuning Versus LoRA 44
7.1.3 Training Behaviour 44
7.1.4 Performance by Language 45
7.1.5 Errors and Triage 45
7.1.6 Choice of Model for the Framework 46
7.1.7 Limitations 47
7.2 Blockchain Layer 47
7.2.1 Ledger Performance under Hyperledger Caliper 47
7.2.2 End-to-End Verification 49
7.3 POPIA Alignment 49
7.3.1 Design Choices 49
7.3.2 Open Questions 50

8 Conclusion & Future Work **50**
8.1 Conclusion 50
8.1.1 Answers to the Research Questions 51
8.1.2 Overall 52
8.2 Future Work 53
8.2.1 Multi-Host and Inter-Institution Deployment 53
8.2.2 Real Election Data and User Evaluation 53
8.2.3 Identity, Revocation and Retention 53
8.2.4 Record Integrity, Verdict Model and Voting Performance 54

1 Introduction

In the months before South Africa's 2024 general election, AI-generated deepfakes in which Joe Biden, Donald Trump and the rapper Eminem appeared to intervene in the campaign spread through politically active WhatsApp groups [1]. Gondwe [1] calls the 2024 elections a watershed moment for misinformation research on the continent. When group members posted fact-checks, the corrections were often met with suspicion, dismissed as propaganda, or reframed as a foreign attempt to meddle in domestic politics.

Those observations frame this study. False content now travels faster than any single organisation can verify it, the corrections that do appear are not trusted by the people who most need them, and the automated tools that could help with scale were not built for South Africa's languages. This report presents a conceptual framework that addresses all three together, and evaluates it through a working prototype. The framework combines a multilingual AI model that flags likely misinformation in English, Sepedi and isiZulu with a permissioned blockchain on which a consortium of fact-checking organisations records, votes on and publishes its verdicts in a form that no single member can alter. Following Wardle and Derakhshan, as summarised by Monnier and Ségur [2], disinformation is false content created to deceive and misinformation is false content shared without that intent, because a monitoring system rarely knows intent, this report uses misinformation as the umbrella term.

The institutional response to the 2024 election was substantial. The Electoral Commission of South Africa (IEC), Media Monitoring Africa, the major platforms, newsrooms and a civil-society fact-checking coalition all took part [3], [4], although the media had to cover the election with limited resources [5]. The result is an ecosystem of many capable actors, each keeping its own record of what it has checked and concluded. What it lacks is a shared, verifiable record that connects those conclusions. This study argues that this gap limits the ecosystem as much as any shortage of detection technology, with the 2026 local government elections, held on 4 November 2026, as its immediate context.

Fact-checking itself works: in simultaneous experiments in Argentina, Nigeria, South Africa and the United Kingdom, fact-checks reduced belief in misinformation, and most of the effects were still visible more than two weeks later [6]. The weakness lies in the system around it. It is reactive, with the same zombie claims debunked every two years [7]. It is fragmented, with verdicts published on different rating scales, so that establishing whether two organisations agree has taken a dedicated research project rather than a simple query [8], [9]. And it struggles to earn trust, as fact-checks are

dismissed as propaganda or seen as partisan [1], [7], [10].

Automation is unavoidable at election scale, but in South Africa it runs into language. Social media platforms do not adequately moderate non-English language content [11], and in a review of misinformation detection research, 67% of studies used English datasets and fewer than 10% focused on low-resource languages [12]. South African studies have begun to close the gap in Tshivenda, Sesotho, Sepedi and isiZulu [13–16], but each machine-translated an English dataset and none addresses elections. Nor can a classifier be the final word since an unexplained machine verdict gives a sceptical audience one more reason to distrust. The model in this study therefore acts as a scout rather than a judge, as it labels each claim as likely misinformation or not so that human fact-checkers can prioritise their review, and its label and confidence score are recorded alongside their verdicts, not in place of them.

Because the core problem is that independent organisations cannot easily share, compare and trust each other's verdicts, the remedy must be organisational as well as technical. A permissioned blockchain gives all member parties a unified view of the data, while no single party can change it without the consensus of the others [17]. A ledger is not a source of truth, so in this framework human fact-checkers, not the model, stand between a claim and its final status on the ledger. The ledger must also respect the Protection of Personal Information Act (POPIA), recording decisions about content without becoming a permanent register of the people who wrote or shared it.

These topics of research seldom meet. Detection studies stop at the classifier, fact-checking research proposes organisational rather than infrastructural remedies and blockchain designs for news verification are not built for low-resource languages or for verdicts shared among many organisations. No work reviewed in this study combines detection in South African languages, verdicts reached jointly by multiple independent organisations and a tamper-evident shared ledger, or measures how such a system performs as the number of member organisations grows.

The report is organised as follows. Section 2 sets out the background and Section 3 reviews related work. Section 4 states the problem, the research questions and the five requirements derived from the literature, and Section 5 presents the conceptual framework that meets them, the central contribution of this study. Section 6 describes the research method and the prototype that instantiates the framework, Section 7 evaluates the prototype, and Section 8 concludes.

2 Background

This section sets out the evidence from which the framework's requirements are derived: the electoral threat (Section 2.1), the fact-checking ecosystem that responds to it (Section 2.2), the state of multilingual AI detection (Section 2.3), blockchain as a shared record (Section 2.4) and the constraints of POPIA (Section 2.5). Section 4.5 draws these together as five requirements.

2.1 Election Misinformation in South Africa

South Africa's electoral law already anticipates false political communication. Section 89(2) of the Electoral Act 73 of 1998 prohibits publishing false information intended to disrupt an election or influence its outcome, and Item 9 of the Electoral Code of Conduct prohibits parties and candidates from publishing false or defamatory allegations [4], [18]. These provisions were designed for identifiable speakers using rallies, print and mass media, and even then the line between false fact and legitimate opinion was contested: Democratic Alliance v. African National Congress turned on whether a campaign SMS sent to more than 1.5 million voters was a statement of fact or of opinion [18].

The 2024 national and provincial elections changed this environment. The African National Congress (ANC) lost its national majority for the first time since 1994, the six-month-old uMkhonto weSizwe Party (MKP) won about 15% of the vote, and a ten-party government of national unity followed [19]. Mare describes 2024 as the first South African election in which TikTok and AI-generated deepfakes took centre stage, amid declining trust in the media and the rise of influencers [20], and Allen and le Roux found an active market for online influence in which many influencers were paid to push narratives [19]. Recommendation systems amplified the shift: Facebook's ranking favours content that generates comments and shares, and TikTok's 'For You' page pushes content from accounts users do not follow to large audiences regardless of accuracy [19]. Disinformation was planted well before polling day to prepare supporters to contest the result, portraying the IEC as corrupt and recasting technical glitches and decontextualised clips as proof of rigging, TikTok and WhatsApp videos of IEC boxes in KwaZulu-Natal, for example, were said to show improperly stored ballots, when the boxes held materials for setting up voting stations [4], [19].

Much of this activity was hidden. WhatsApp has the largest share of South Africa's social media users, but its encryption makes it difficult to research, and senior editors worried that vernacular WhatsApp groups did more to spread false information than X and Facebook [19]. In 22,384 messages

from 47 politically active WhatsApp groups, Gondwe found that disinformation relied on fear-based appeals (41%), identity-based rhetoric (32%) and content imitating credible journalism (27%), and spread because it struck an emotional chord among like-minded members, not because it was accurate [1].

Generative AI added synthetic media, from a video in which Donald Trump appeared to endorse the MKP, labelled fake by Africa Check only after it had spread widely on X, to a voice-cloned Joe Biden warning of sanctions [1], [19], replicated voices and images of public figures defeated static security arrangements [4]. Allen and le Roux, focusing on Facebook and X, judged deepfakes limited [19], whereas Gondwe found WhatsApp groups saturated with them [1], which shows how invisible synthetic media in closed groups is to open-platform monitoring. Only about 2% of detection studies examine audio [12], and generative tools produce convincing but hallucinated content [21]. Allen and le Roux recommended that the IEC and the Government Communication and Information System (GCIS) prepare jointly for the 2026 local government elections [19], held on 4 November 2026. Because local elections are contested at ward and municipal level, they shift misinformation towards more localised and linguistically diverse content spread through community messaging groups, the conditions under which the 2024 evidence shows monitoring to be weakest [1], [19].

2.2 The Fact-Checking Ecosystem: Fragmentation and Trust

The response to election misinformation involves many actors: the IEC, which worked with Media Monitoring Africa (MMA) on the Real411 complaints platform and signed a Framework of Cooperation with Google, Meta, TikTok and MMA [4], [18], the platforms' own fact-checking partnerships [11], a GCIS network of computer scientists, fact-checkers and think tanks [19], the South African National Editors' Forum (SANEF) [4] and fact-checkers led by Africa Check, the continent's first independent fact-checker [7], alongside News24's disinformation desk, the Daily Maverick's Factcheck Hub [3] and a 2024 election coalition including AFP Fact Check, the Mail & Guardian, Caxton Media and SABC News [4].

These initiatives are valuable, but they are organised around individual organisations, platforms and funders rather than shared verification infrastructure. A claim may be assessed by one organisation, labelled on one platform and reported through a separate channel without the outcome being visible to the others. Fact-checkers are not routinely present in South African newsrooms [19], they describe their work as resource-intensive and

reactive [7], and in encrypted WhatsApp groups their efforts ran in parallel rather than in response to misinformation [1]. Dependence on platforms is a further exposure: in January 2025 Meta ended its third-party fact-checking programme in the United States [22]. One South African fact-checker described debunking the same claims every two years, and such recurring zombie claims drain resources [7], which shows the value of a persistent, shared record of earlier verdicts. None of the South African sources reviewed describes a common record through which the IEC, fact-checkers, media and civil society can see which claims have been assessed, by whom, on what evidence and with what outcome.

International evidence locates the weakness in how fact-checks are produced rather than in what they conclude. Markowitz et al. found 22.6% disagreement between The Washington Post and PolitiFact in which statements they selected, but nearly complete agreement on bottom-line veracity [8]. Lee et al. found that 30.4% of claims checked by both Snopes and PolitiFact received diverging ratings, almost all of which disappeared once differences in rating scales were adjusted, and that only 6.5% of their articles concerned matching claims [9]. Because selection criteria and rating schemes are not standardised, outputs are hard to compare or audit, and differences in selection can look like bias. Fact-checkers outside the Western world try to check both sides of the partisan divide equally to satisfy partisan audiences [23], South African fact-checkers report a neutrality bias that can obscure the truth [7], and the true/false verdict itself struggles with complex situations [2], as does LLM-based fact-checking with nuanced labels [24].

Fact-checking also depends on public trust, which is under strain. Trust in news in South Africa fell from 61% in 2022 to 55% in 2025, and 73% of respondents worry about telling real from fake news online, although the sample over-represents younger, English-speaking users [3]. South African fact-checkers are increasingly seen as partisan or accused of lying [7], and distrust of fact-checking services is often grounded in perceived political bias [2], [8], [10]. In the 2024 WhatsApp groups, corrections were written off as propaganda or foreign interference because they came from outside the group's epistemic enclave [1], and survey evidence suggests that platform fact-checks barely deter sharing [25]. The IEC, itself accused of favouring the ANC and of having been bought [4], [19], was slow to explain technical glitches, leaving a vacuum in which rumour flourished [19].

A single authority issuing verdicts on contested political claims is therefore likely to attract the same allegations of bias. Brandtzaeg and Følstad argue that fact-checking services should be transparent about their process and could act as platforms for collaborative verification [10], and groups identify false posts more accurately than individuals [26]. Verdicts should therefore

come from several independent organisations, and the record should show who verified what, on what evidence and when, so that criticism is directed at verifiable decisions rather than at presumed motives.

2.3 AI for Multilingual Misinformation Detection

Early misinformation detection used hand-crafted features or TF-IDF representations with classifiers such as logistic regression, support vector machines and naive Bayes [12], these are simple to train and interpret but capture little context. Transformer models use attention to represent relationships between tokens, and about 81% of the studies reviewed by Rananga et al. preferred deep learning [12]. Fine-tuned BERT-like encoders generally outperform decoder-only LLMs on classification, while LLMs are more robust to perturbed text [27], and in fact-checking with Llama-3 models, evidence retrieval improved performance but nuanced labels remained difficult [24]. Detection is nevertheless not the same as establishing truth such that a claim's factual status may depend on its source, timing, surrounding events and evidence [2], [24], generative models can produce unsupported output [21], and explainability in automated fact verification lags the wider field [28]. In Gondwe's study, an AI tool not validated for African languages had its classifications corrected by bilingual coders [1]. A classifier can therefore prioritise content for investigation, but its output should not be treated as a final judgement.

Language is the main obstacle in South Africa. English datasets were used in 67% of the misinformation detection studies reviewed by Rananga et al., and fewer than 10% focused on low-resource languages, which lack labelled data, benchmarks, pretrained models and tools [12]. South Africa has 12 official languages, but platforms moderate non-English content inadequately: Meta's fact-checking network covered five of them in 2024, and TikTok took no action against degrading content in South African vernacular whose English equivalents it banned [11]. South African studies have begun detection in Tshivenda, Sesotho and Sepedi, each by translating an English dataset because native labelled data is scarce, and translation quality strongly affects performance [13], [14], [15]. Data augmentation and pretrained transformers have been explored here and in other low-resource languages such as Bangla [15], [29].

Real South African text is also code-switched and informal. About 30% of tweets in five South African languages mix in English [30], code-switching, unstandardised writing and scarce annotation make detection harder [12], [31] and zero-shot GPT-4 performed far worse than a fine-tuned

Afrocentric model on such text [30]. Multilingual models such as AfroXLMR, SERENGETI and the Zabantu models now cover South African languages by transferring representations from better-resourced languages [12], [15], [31], but performance varies by language: AfroXLMR performed better on isiXhosa and isiZulu than on Sepedi, Setswana and Sesotho, and an ensemble reached weighted F1-scores above 77% on the Nguni languages but above 63% on the Sotho-Tswana group [31]. Detection for South Africa must therefore be evaluated language by language, and ultimately must handle multimodal content: about 73% of reviewed studies used text and only 2% audio, although voice-cloned and decontextualised recordings circulated in 2024 [1], [12], [19].

2.4 Blockchain as a Shared, Tamper-Evident Record

Verification outcomes are currently held in systems controlled by individual organisations: a fact-checker's website, a platform's labels or a single complaints portal. Centralised trust models concentrate control and create single points of failure, and their lack of openness means their ratings are not universally accepted [32], centrally stored news data can be altered without its owners' knowledge [33] and centralised evidence storage is poorly traceable and open to tampering [34]. The 2024 elections showed how this feeds misinformation. When the IEC's results website went offline for more than an hour, claims that delays were deliberate and that the system was being rigged gained traction [4], [20], alongside reports of malfunctioning voter management devices [35]. The results themselves were protected because copies were posted at every polling station and given to party agents, which observers commended [35], and parties lodged 579 formal objections through established channels [4]. Fact-checking records have no equivalent of those results slips: a record held by one organisation can be edited or removed without others being able to tell, a sensitive concern given South Africa's history of censorship [36].

Blockchain systems maintain a shared ledger whose transactions are replicated or validated across participants [37], so that no single entity can change the data on its own [38] and parties who do not fully trust one another can agree on a common set of facts [17]. Blockchain has been proposed for news traceability and for the provenance of online content [39], [33], [40] (Section 3.3). Its relevance here is provenance and auditability, not classification: immutability is useful only if what is entered is accurate, information from outside the chain creates a first mile/last mile problem, and where transactions are hard to verify, human judgement is needed before

data is recorded [17]. A ledger can preserve a verification event, with its identifier, timestamp, status and a cryptographic reference to the material [39], [33], [41], while AI supports the identification of content and people decide.

Permissioned blockchains restrict participation to identified, authorised entities under defined governance [37], which is why governments adopting blockchain usually choose them [38]. Hyperledger Fabric supports confidentiality between organisations through certificates that establish each organisation's legitimacy, channels and off-chain storage of transaction details [42], which suits a consortium in which the IEC, fact-checkers, media and civil society share one verification record. As in the insurance consortium of IBM, AIG and Standard Chartered, members must cooperate but none should own the record, and governance remains essential, because trust between members still manages the first mile/last mile problem [17]. A ledger can also record AI decisions and the provenance of their data, so that they can be audited [43].

2.5 POPIA and the Immutability Paradox

Any system that records claims, evidence and verification activity in South Africa must comply with the Protection of Personal Information Act 4 of 2013 (POPIA), whose main provisions took effect on 1 July 2020 [44], [45], [46]. Processing needs a lawful basis, such as consent, legitimate interests or, for a public body such as the IEC, a public law duty and data subjects may object (section 11). Records may not be kept longer than their purpose requires and must then be destroyed or de-identified so they cannot be reconstructed (section 14), data subjects may request correction or deletion of inaccurate, excessive or unlawfully obtained information (section 24), responsible parties must secure integrity and confidentiality (section 19) and information de-identified beyond re-identification falls outside the Act (section 6) [44], [45]. Election misinformation records are likely to contain personal information, and both political persuasion and alleged offences, such as the bribery and rigging that election disinformation often alleges [4], are special personal information whose processing needs a specific authorisation (sections 26 and 27) [44]. The journalistic exclusion (section 7) may not extend to infrastructure shared by the IEC and civil society and any restriction on spreading information, true or false, must be justified under the Constitution's protection of free expression [44], [18].

These duties collide with a blockchain's permanence. Records are append-only and replicated [17], [38], whereas POPIA requires correction and dele-

tion [44]. Finck observes the same conflict under the GDPR [47], Almiani et al. report that it has pushed implementations towards permissioned or hybrid models [38] and South African public-sector work raises the concern that identities on a ledger could compromise POPIA [48]. Fact-checkers also correct their own work [7], so a verification ledger must allow correction without erasing the history that gives it evidential value. Because personal data written on-chain is copied to every participant and cannot be destroyed by any one of them [37], [47], the design question is which data are recorded immutably.

Hybrid architectures answer it by keeping substantive data off-chain and recording only integrity information on-chain [49], [50]. A cryptographic hash changes whenever its input changes, so a later check can confirm that content matches its recorded hash [34], [41], and personal data held off-chain and linked only by a hash can be corrected or erased while the ledger still shows that a verification took place [47]. Fabric's channels and Identity Mixer add privacy controls [42], and zero-knowledge proofs and related techniques could prove that a verification met an agreed rule without revealing personal data [38]. The compliance question remains open at the edges: metadata can reveal information even when hashed [38], and a hash of a short or predictable value can be re-identified by hashing candidates, so a remaining hash may itself be personal information [44], [47]. A POPIA-aligned design must therefore treat on-chain metadata as potentially personal and keep it to a minimum.

3 Related Work

This study draws on two bodies of work that have so far developed separately: the automatic detection of misinformation in South Africa's lower-resourced languages, and the use of permissioned blockchains to keep tamper-evident records shared by institutions that do not fully trust one another. A smaller third body of work proposes shared infrastructure, including blockchains, specifically for fact-checks and fake news. Sections 3.1 to 3.3 review each in turn.

3.1 Misinformation Detection in Low-Resource South African Languages

3.1.1 South African Detection Studies Built on Translated Data

Rananga et al.'s review calls for new low-resource datasets, data augmentation, transfer learning and the fine-tuning of multilingual transformers such as XLM-R [12], and recent South African studies have acted on it, each machine-translating an existing English dataset into the target language. Mokoena et al. translated the CoAID COVID-19 dataset into Sesotho with two systems and found that the same SVM scored an F1 of 0.908 on the better translation against 0.866 on the other [14]. Manaswe et al. translated 1,500 sentences of CoVID19-FNIR into Sepedi and tested six augmentation techniques: classical models scored highly, while the fine-tuned Zabantu encoder reached at most 79% accuracy [15]. Sethosa et al. applied the same strategy to hate speech in isiZulu, but found that translation struggled with offensive text and cultural context [16]. De Jager et al. built the first South African misinformation dataset from Real411 complaints and found that adding local samples to a multimodal BERT and ResNet model significantly improved detection, although Real411 is predominantly English [51].

These studies show that detection in Sesotho, Sepedi and isiZulu is feasible without native corpora, that translation quality matters, and that locally relevant content improves detection. None addresses elections, however, and most rely on classical TF-IDF classifiers. This study keeps the translation step but changes its source (Section 6.2), and retains TF-IDF baselines for comparison (Section 6.3).

3.1.2 Language Models for African-Language Classification

Mabokela et al. [30] found that fine-tuned AfroXLMR outperformed generative LLMs such as GPT-4 and BLOOM on sentiment classification of tweets in five South African languages, including Sepedi and isiZulu, and attribute its advantage to pre-training on African-language text that copes with agglutinative morphology and English code-switching. For English fake news, Raza et al. [27] found that fine-tuned BERT-like encoders outperformed fine-tuned decoder-only LLMs on every classification metric at lower cost, while the LLMs were more robust to perturbed text, and that AI-generated labels checked by people produced better classifiers than labels inferred from the news source. Zhang et al. [52] likewise found that encoders generally outperform LLMs on classification with fewer computational resources, although without fine-tuning the LLMs, which did better on knowledge-intensive tasks

such as hallucination detection, arguably the task type closest to fact verification. These comparisons inform the choice of models in Section 6.3.

3.1.3 LLM-Generated Training Data

Nadăş et al. [53] survey LLM-generated training data for settings where labelled data is scarce, expensive or sensitive, one study they review found gains of 3–26% in accuracy or F1-score when 100 synthetic samples were added to 100 real ones. They recommend topic-controlled or randomised prompting to increase diversity, and filtering and human validation because generated text can contain factual errors, may not match the style and distribution of real data, and can amplify bias. The prompt axes and the back-translation check with human review in Section 6.2 follow these recommendations.

3.1.4 Parameter-Efficient Fine-Tuning

Shuttleworth et al. [54] examined whether Low-Rank Adaptation (LoRA) and full fine-tuning learn equivalent solutions. Even when both reached the same task accuracy, their weight matrices differed: LoRA introduced new high-ranking singular vectors, which the authors call intruder dimensions, that made the model forget more of its pre-training distribution and accumulated when tasks were learned in sequence. Setting the LoRA scaling factor to twice the rank ($\alpha = 2r$) produced fewer intruder dimensions and better generalisation than a smaller α. Section 6.3 therefore compares the two methods directly rather than assuming they are equivalent, and sets $\alpha = 2r$.

3.2 Permissioned Blockchains for Shared Institutional Records

The blockchain layer of this study records the decisions of organisations that do not fully trust each other, keeps documents off-chain and gives organisations different rights. Studies in health care, criminal justice, public administration and supply chains, where institutions also share records that may later be disputed, have established most of the design patterns it adopts.

In health care, Tith et al. built a Fabric consortium through which hospitals share records, with identities issued by each member's certificate authority, only addresses and hashes on the ledger, and every access logged [55], a review of 57 Fabric health studies found the platform often combined with IPFS and benchmarked with Caliper, but seldom evaluated under real conditions [42]. Criminal justice is the closest institutional analogue, since records

are shared among agencies with separate mandates and must withstand later scrutiny. ShareCrimE, a Fabric network with Raft ordering for six South African criminal justice entities, is the nearest local precedent, but it stored case data in the world state and was evaluated only by simulation [56]. Amin et al. showed with Caliper that Fabric handles crime-reporting workloads at sub-second latency [57], and ChainSEAL anchors only the SHA-256 hash, CID and chain-of-custody metadata of evidence stored on IPFS, because storing even a 1 MB file on-chain would exceed Ethereum's block gas limit many times over [34].

In public administration, Alotaibi et al. used design science research, the approach this study follows, to build GovBlockchain, a Fabric model for transparent procurement records with different views for auditors and citizens [58], and other studies anchored hashes of e-government documents and academic credentials so that off-chain documents can be verified later [59], [60]. Zhu et al. formalised the on-chain/off-chain pattern, saving about 75% of storage while noting weaker protection and availability off-chain [50], and Dai et al. checked permissions inside Fabric chaincode at 214 transactions per second, with latency strongly affected by node placement [61].

Across these domains a common design emerges, and this study adopts it: a consortium of known institutions, each issuing identities through its own certificate authority, Raft ordering, documents kept off-chain, with only their hash or content identifier on the ledger, separate roles for submitting, deciding and reading, and Hyperledger Caliper for measuring performance. Where these systems enforce roles on the ledger, however, the roles are bound to Ethereum account addresses [34], [59] or to role records in the ledger state [61] rather than to certificate attributes that every endorsing peer can check (Section 6.5.2), and read-only access for external parties is conceptual [58] or enforced outside the ledger [62], as it also is in this study, where the observer role is enforced by the gateway (Section 6.7).

3.3 Shared Fact-Checking Infrastructure and Blockchain Approaches

Shared infrastructure for fact-checks already exists. ClaimReview markup lets tools such as Google Fact Check Explorer index fact-checks from around the world [63], and the International Fact-Checking Network's CoronaVirusFacts/DatosCoronaVirus Alliance pooled fact-checks from more than 70 countries during the COVID-19 pandemic [64]. These resources make published verdicts easier to find, but they collect verdicts that each organisation reaches on its own, are not tamper-evident and do not produce joint verdicts.

Huckle and White built Provenator, an Ethereum application that records provenance metadata so that anyone can check a media file's hash against its creator's record, motivated by a photograph of ballot boxes reused as false evidence of vote rigging [39]. A hash matches only an identical file, and, the authors argue, technology can verify the origin of a resource but not the truth of a story, which takes human skills, this study's ledger likewise proves the integrity of a report, while human fact-checkers decide its verdict. Qayyum et al. proposed, without implementing, smart contracts that enrol and revoke news publishers with reputation scores, naming AI-based trustworthiness assessment as future work [40] and Wang et al. kept news data off-chain with on-chain polynomial commitments, analysed formally but not evaluated empirically [33].

Farhan et al. proposed VeriTrust, the design closest to this study, which combines self-sovereign identity, Merkle-tree anchoring of content hashes on an Ethereum-compatible chain, IPFS storage, AI flagging, credential-based voting by fact-checkers and a reputation system, with no personal data on-chain [49]. It sets out the sequence this study implements: automated flagging, human verification and a tamper-evident record. However, it is conceptual, with performance projected from published benchmarks, its token rewards and stake-based voting suit an open network rather than a consortium of known institutions and its authors acknowledge gas costs and the conflict between immutability and erasure rights, which Sections 6.5.1 and 6.6 address. Kuznetsov et al. note that a ledger can record the provenance of a model's training data and versions, but that combining AI and blockchain also brings risks, from adversarial attacks to key management [43], this study records a model version string with each report (Section 6.6.1).

These studies show that a ledger can make decisions about misinformation tamper-evident and auditable, and VeriTrust shows how automated detection and human verification can be organised around one. None, however, has been implemented with measured ledger performance, none is designed for a permissioned consortium of known institutions with certificate-based identities, and none considers African languages, elections in Africa or a specific data protection law such as POPIA.

3.4 Summary and Research Gap

Taken together, the reviewed work makes this research feasible: translated training data, Afrocentric encoders, LLM-generated corpora and Fabric consortia with Raft ordering, off-chain storage and hash anchoring have each been demonstrated. No study found in this review, however, brings these

elements together. In particular:

(i) No labelled election-misinformation corpus or detector was found for Sepedi or isiZulu. The South African detection studies address COVID-19, hate speech and general misinformation, and build their data by translating English datasets that contain no South African election content.

(ii) The only design that links automated flagging to human verification on a ledger [49] is conceptual and targets an Ethereum-compatible network with token incentives, not a consortium of known institutions.

(iii) No reviewed system binds roles to attributes in participants' X.509 certificates evaluated inside chaincode.

(iv) No reviewed system decides what to store on-chain with reference to POPIA.

(v) No misinformation-specific system reports measured latency or throughput. The measurements available in related domains [57], [61] come from systems without a detection stage.

This study addresses each of these gaps with the artefact described in Section 6, from the election-specific corpus and AfroXLMR detector (Sections 6.2 and 6.3) to the certificate-based Fabric consortium, its public verdict portal, its end-to-end functional test and its Caliper benchmarks (Sections 6.4 to 6.8).

4 Problem Statement and Requirements

4.1 Problem Statement

Election misinformation in South Africa now spreads through open platforms, encrypted messaging groups and AI-generated media faster than any single organisation can verify it. Fact-checkers, newsrooms, the Electoral Commission of South Africa (IEC) and civil society each respond, but they keep separate records, apply different rating schemes and are each exposed to accusations of bias. Automated detection could help with scale, but most of it has been built for English, and labelled data in South African languages is scarce. No existing approach combines multilingual detection with a shared, tamper-evident and privacy-aware record of how independent organisations verified

each claim. The research problem is therefore the absence of an integrated AI and blockchain framework that supports multilingual misinformation detection together with transparent, accountable and trusted fact-checking in South Africa's multi-stakeholder electoral context.

4.2 Research Question

How can an integrated AI and blockchain-based framework be designed and evaluated to enable tamper-resistant, transparent, multilingual monitoring and verification of election misinformation within South Africa's multi-stakeholder electoral context?

4.3 Sub-Research Questions

1. **Artificial Intelligence enabled multilingual misinformation detection.**

   Which artificial intelligence enabled approaches are most suitable for detecting misinformation across multiple South African languages, and how can this be adapted to address low-resource data?

2. **Blockchain architecture to enable transparent, tamper-resistant, and auditable fact-checking.**

   Which blockchain architecture best balances data integrity, privacy, accessibility and scalability for a consortium-based fact-checking network in South Africa?

3. **Artificial Intelligence enabled misinformation detection and blockchain integration.**

   How can artificial intelligence enabled misinformation detection be integrated with a blockchain ledger to support an end-to-end, tamper-resistant and auditable monitoring workflow?

4. **Privacy and data-protection alignment.**

   Which privacy-preserving and cryptographic techniques can support alignment with South Africa's Protection of Personal Information Act (POPIA) while maintaining the transparency and auditability of fact-checking decisions?

5. **Evaluation of technical properties.**

   To what extent can the integrated AI-blockchain conceptual framework demonstrate the technical properties on which transparency, traceability and auditability depend in multi-stakeholder misinformation monitoring?

4.4 Research Objectives

The following objectives were pursued, each addressing one or more of the sub-research questions above, referred to as SRQ1 to SRQ5. Sections 7 and 8 report the extent to which each was met, and Table 17 records the status of each.

Objective 1. To examine the challenges of detecting misinformation within multilingual, low-resource South African language contexts (SRQ1).

Objective 2. To investigate existing AI/NLP approaches for misinformation detection and assess their suitability for a multilingual South African election environment (SRQ1).

Objective 3. To design a blockchain architecture that balances data integrity, privacy, accessibility and scalability for a consortium-based fact-checking network (SRQ2).

Objective 4. To integrate the AI-based misinformation-detection component with the blockchain ledger to form an end-to-end, tamper-resistant and auditable monitoring workflow (SRQ3).

Objective 5. To implement a privacy-preserving on-chain/off-chain data model aligned with South Africa's Protection of Personal Information Act (SRQ4).

Objective 6. To implement a working prototype demonstrating the technical feasibility of the integrated framework.

Objective 7. To evaluate the prototype against measurable criteria: AI detection quality (accuracy, precision, recall, F1), blockchain system performance (throughput, latency, scalability) and POPIA alignment (SRQ5).

Objective 8. To analyse the limitations, trade-offs and governance considerations of integrating AI and blockchain within a multilingual election misinformation monitoring system.

4.5 Requirements

The background in Section 2 shows that election misinformation in South Africa is networked, algorithmically amplified, partly hidden in encrypted groups, increasingly synthetic and often directed at the IEC itself [1], [4],

[19], [20], and that local government elections, contested at ward level, push these threats towards a more localised form. Together with the gaps identified in Section 3, it establishes five requirements for an election misinformation monitoring system:

- **R1 Multilingual processing with scarce data.** The system must process multilingual content in a language environment where labelled resources are limited [12], [13], [14], [15], [30], [29], [31].
- **R2 Classification that supports verification.** Automated classification should support rather than replace contextual verification, because classification alone cannot establish the factual status of every political claim, and because fact-checking organisations differ in how they select and rate claims [2], [8], [9], [24].
- **R3 A shared record with no single custodian.** Verification records must be shared across organisations that do not fully trust one another, and must not depend on a single custodian whose failure or alleged partiality could undermine the record [4], [7], [17], [32].
- **R4 Traceable decisions.** The verification process must be transparent and traceable, so that allegations of bias can be tested against a record of who verified what, using which evidence and when [1], [10], [39], [33], [40], [41].
- **R5 Compliance with POPIA.** The system must comply with POPIA, which requires that personal information can be corrected and deleted, so the conflict with ledger immutability must be resolved through the architecture rather than ignored [44], [38], [47].

Section 5 presents a framework that meets these requirements, and Section 8.1.1 sets them against the evaluation evidence (Table 16).

4.6 Scope and Limitations

The framework is evaluated through a prototype under controlled conditions. The detection component is trained and evaluated on a synthetic, machine-translated corpus rather than live social media content, covers English, Sepedi and isiZulu, and handles text only. The ledger is benchmarked with up to twenty organisations, each with its own peer and certificate authority, on a single host, deployment across physically separated institutional servers is future work. Verdicts are binary, and a closed report cannot be corrected

on the ledger. The evaluation targets detection quality, report submission and full-ledger reads. Usability and user acceptance of the web client, voting under load, the throughput of the AI pipeline and web client, an experimental comparison with a centralised database, and effects on public trust or collaboration between institutions are outside its scope.

4.7 Assumptions

The design and evaluation rest on the following assumptions:

- **Known participants.** Consortium members are vetted institutions that may fail or disagree but do not collude to rewrite the ledger, so the ordering service's threat model is crash faults rather than Byzantine behaviour (Section 6.6.2). Chaincode rules limit what a member acting in bad faith can do through the contract, but because any two thirds of organisations can endorse a transaction, the two-thirds rule depends on no set two thirds of members colluding at peer level.
- **Key custody.** Each organisation protects its own private keys, the single-host prototype holds all identities in one service (Section 6.4.1).
- **Good-faith review.** Fact-checkers vote according to the evidence, the ledger records which organisation voted and how, not whether a vote is correct.
- **Triage, not verdicts.** The model's output is advisory as a claim's status is never decided by the model, only through member organisations' votes and officials' closing actions (Section 6.6.2).
- **Relative performance.** Benchmarks on one host show how the ledger behaves as the consortium grows, absolute figures will differ on separate institutional servers.

5 Conceptual Framework

This section presents the study's central contribution: a conceptual framework for monitoring election misinformation across languages and across independent organisations. The framework meets the requirements in Section 4.5 through complementary functions. An AI component addresses the analytical challenge of identifying potentially misleading multilingual content, a permissioned shared ledger addresses the institutional challenge of

preserving verification records across competing stakeholders and an off-chain document store, linked to the ledger by cryptographic references, reconciles that record with POPIA. Human verification sits between the model and the ledger, so that the ledger never certifies the truth of the model's output and the first mile/last mile problem is handled before information is committed [17]. The framework is stated independently of particular technologies: Section 6 describes one instantiation, and Section 7 evaluates it.

5.1 Design Principles

The framework rests on one separation of responsibilities: the AI model advises, people decide jointly and the ledger records. A model can process claims at election scale but cannot establish whether a political claim is true (Section 2.3). Independent organisations can reach defensible verdicts but have no shared record of them (Section 2.2). A ledger can show that a record has not been altered but not that it is correct (Section 2.4). Each component therefore does only the part of the work it is suited to, which gives five design principles:

- **P1 Advisory automation.** The model labels every claim and attaches a confidence score, but never closes, removes or decides a claim, its output only orders the review queue (R1, R2).
- **P2 Joint verdicts.** A claim's status is decided only by the votes of independent member organisations under a published threshold, so no organisation, including the one that submitted the claim, decides alone (R2, R3).
- **P3 A shared, tamper-evident record.** Every report, vote and status change is written to a ledger replicated across the members, which none of them can alter unilaterally (R3, R4).
- **P4 Minimal on-chain data.** The ledger holds only hashes and structured metadata, documents that may contain personal information stay off-chain, where they can be corrected or removed (R5).
- **P5 Replaceable components.** The model, the ledger and the document store meet only at narrow interfaces (a label, a confidence score and a model version, a hash and a document identifier), so each can be replaced without changing the others or the verification rules (R1, R3).

5.2 Actors and Roles

Four kinds of actors take part, each with a defined scope of action:

- **Public.** Submits claims anonymously and reads closed verdicts with their votes, reasoning and evidence, claims still under review are not shown.
- **Member organisations.** Fact-checkers, newsrooms, the IEC and civil-society monitors that form the consortium. Each acts through officials, who govern membership and close reports after the review window, and through fact-checkers, who submit claims and cast the organisation's single vote on each report.
- **Observers.** Auditors within the consortium with read-only access to the full record.
- **Automated services.** The detection model and the services that carry its labelled claims to the ledger, which may submit reports but never vote.

Roles are bound to each organisation's identity on the ledger, so that permissions are enforced where the record is kept rather than by any one organisation's software (Section 6.7).

5.3 Claim Lifecycle

A claim passes through five stages (Figure 1):

- **Intake.** Claims arrive anonymously from the public through a tip line, or in bulk through an authenticated ingestion interface.
- **AI screening.** A multilingual classifier labels every claim as likely misinformation or not and attaches a confidence score. The label is advice: it helps reviewers prioritise and does not close or remove a claim.
- **Anchoring.** The claim and its label are stored off-chain as a content-addressed document, and a report holding the document's hash, the label and a review deadline (72 hours in the prototype) is written to the ledger.

- **Joint verification.** Each member organisation reviews the claim without seeing how the others voted and casts one vote on the record. A claim is confirmed as misinformation when two-thirds of the registered organisations agree.
- **Publication and audit.** Once a report closes, its outcome is published with each organisation's vote, reasoning and evidence, the transaction history and an integrity check. Claims still under review are not published, so the framework does not amplify them.

A report that can no longer reach the threshold is closed as not confirmed rather than as accurate, and one still open when its review window ends may be closed by a fallback rule or expire, so that no claim stays open indefinitely (Section 6.6.2).

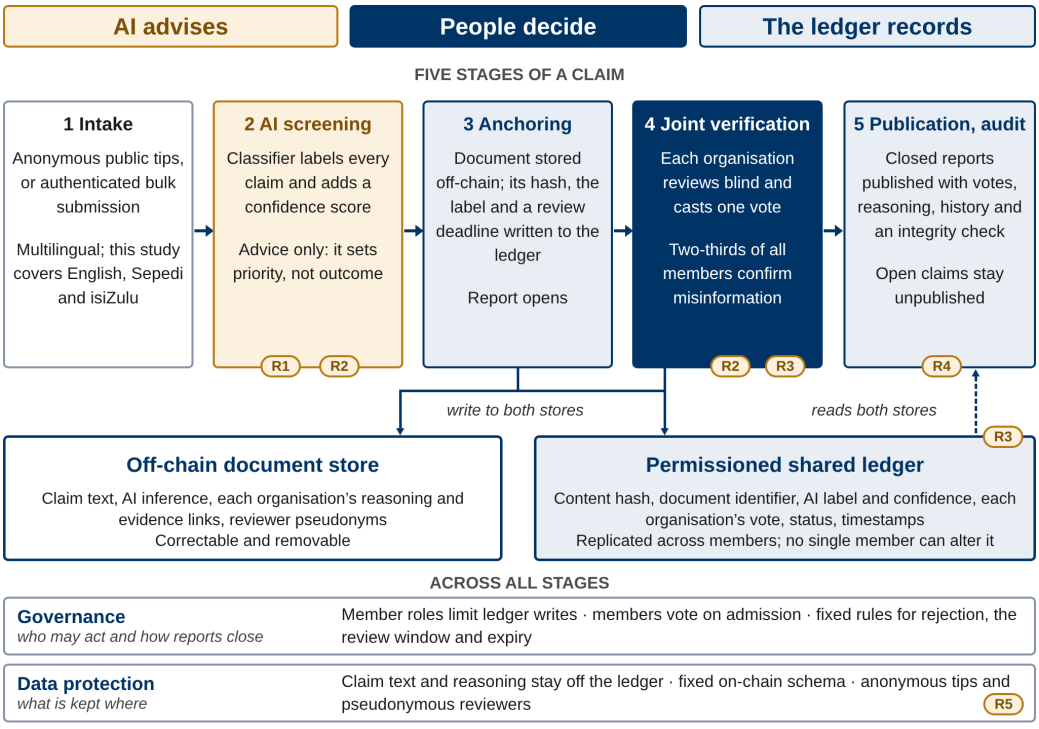


Figure 1: Conceptual framework. R1 to R5 mark the requirement from Section 4.5 that each element answers. Table 1 gives the prototype's instantiation of each element.

5.4 Governance and Data Protection

Two concerns run across all five stages. Governance determines who may act and how a report closes. Role attributes bound to each member's identity restrict who may write to the ledger, members vote on admitting new organisations under the same two-thirds rule, review is blind, so that each organisation votes without seeing the others' votes and fixed rules govern rejection, the review window and expiry. A two-thirds threshold is used rather than a simple majority because a verdict of misinformation is a public statement on behalf of the consortium and should carry broad agreement, and it guarantees that the submitting organisation can never decide alone.

Data protection determines what is kept where. The ledger holds only hashes and structured metadata in a fixed schema, while documents that may contain personal information stay off-chain, where they can be corrected or removed without altering the ledger. Tips are anonymous, reviewers appear publicly only as their organisation, and claims under review are not published, so the framework neither amplifies unverified claims nor builds a permanent register of the people who wrote or shared them (Section 7.3).

5.5 From Framework to Prototype

The framework does not prescribe technologies: any multilingual classifier, permissioned ledger with identity-bound roles and content-addressed document store that satisfies the principles above can instantiate it. Table 1 shows the instantiation built in this study and the requirement each element answers, and Section 8.1.1 sets the requirements against the evaluation evidence (Table 16).

6 Research Method and Instantiation

6.1 Design-Science Approach

This study followed a design-science research (DSR) approach [65], [66], which suits problems whose objective is to construct and evaluate an artefact that solves an explicit real-world problem: here, the lack of a verifiable, tamper-resistant and privacy-aware audit trail for election misinformation in South Africa. The artefact is the conceptual framework in Section 5, and this section describes the prototype that instantiates it, so that the framework can be shown to be implementable and evaluated against R1 to R5 (Table 1). The evaluation is mixed-methods: quantitative measures of detection

Table 1: Framework elements, the requirements they answer and their instantiation in the prototype.

Framework element	**Req.**	**Prototype instantiation**	**Section**
Multilingual advisory classifier	R1, R2	AfroXLMR-large-76L fine-tuned on a synthetic English, Sepedi and isiZulu corpus, label, confidence and model version for every claim	6.2, 6.3
Intake and screening pipeline	R1, R2	Anonymous tip line and authenticated bulk API, Kafka pipeline with HMAC-signed messages, every claim submitted, flagged or not	6.4
Off-chain document store	R5	Private three-node IPFS swarm holding content-addressed documents	6.6.1
Shared ledger	R3	Hyperledger Fabric consortium with Raft ordering and any-two endorsement	6.5, 6.6.2
Anchoring and integrity check	R4	SHA-256 hash of the claim and AI inference on-chain, verification endpoint recomputes it	6.6.1
Joint verification	R2, R3	Two-thirds threshold enforced in chaincode, blind review in the fact-checking service	6.4.2, 6.6.2
Roles and governance	R3, R4	X.509 certificate role attributes checked in chaincode, admission by two-thirds vote, hash-chained audit log	6.6.2, 6.7
Publication and audit	R4	Public portal with votes, reasoning, evidence, transaction history and integrity check	6.4.2
Data protection	R5	Fixed on-chain schema, off-chain documents and reviewer pseudonyms	6.6, 7.3

quality and ledger performance address Objectives 6 and 7, and a qualitative analysis of governance, auditability and alignment with data protection legislation [44] addresses Objective 5. Sections 6.2 and 6.3 construct the detection component (SRQ1), Sections 6.4 to 6.7 the blockchain component and its integration (SRQ2 to SRQ4), and Section 6.8 sets out the evaluation (SRQ5).

6.2 Synthetic Data Generation

No labelled corpus of South African election misinformation exists in Sepedi or isiZulu. The closest South African studies built their training data by machine-translating an English dataset [13–16], but no existing English dataset covers South African elections. This study therefore generated an election-specific English corpus with a single large language model (LLM), Gemini 3.6 Flash, and translated it into Sepedi and isiZulu, an approach increasingly used where labelled data is scarce, expensive or sensitive [53]. Because every statement is synthetic, no personal information of real social media users was collected. The pipeline ran in three stages (Table 2), and every row of the final corpus can be traced to the prompt and model run that produced it.

Table 2: Stages of the synthetic data pipeline.

Stage	**Tool**	**Output**
1. Prompt construction	Python script, fixed random seed	5,000 prompts, each requesting 10 labelled English samples
2. Generation	Gemini 3.6 Flash	Statements with a label, a claimed source and an explanation
3. Translation	Gemini 3.6 Flash	Each statement in one of English, Sepedi or isiZulu

6.2.1 Prompt Construction and Generation

Prompts were built by a script from a single recorded random seed (797973867), so that the corpus covers the space of election misinformation systematically and can be regenerated exactly. Each of the 5,000 prompts combines one value from each axis in Table 3 and sets four of twelve style features, such as sarcasm, formality, urgency and South African English colouring.

Each prompt asked for ten English samples, five labelled misinformation and five non-misinformation, sharing the prompt's topic, format, setting, tone and style, so that a classifier cannot separate the classes by topic or writing style alone. The prompt required every statement to make a different claim, misinformation to be plausible and written in the voice of someone who believes it, and non-misinformation to be accurate about how South African elections work. Gemini 3.6 Flash generated every statement through the

Table 3: Prompt axes. One value was drawn from each axis for every prompt.

Axis	Values	Examples
Topic	25	voters' roll and registration, special votes, ballot-box chain of custody, coalition talks, load-shedding at voting stations, fake IEC statements
Format	10	WhatsApp forward, transcribed voice note, tweet, Facebook community post, news headline and article, SMS blast, community-radio comment
Setting	8	metro, township, small rural town, village under traditional authority, informal settlement, mining-belt municipality
Tone	8	angry and accusatory, concerned citizen, sarcastic, conspiratorial, neutral news report
Election period	5	2021 and 2024 elections (retrospective), 2026 local government campaign, voting day (4 November 2026), counting and results
Misinformation mechanism	8	fabricated event, real process with false details, true fact with a false explanation, manipulated statistic, false attribution, recycled old incident
Accurate content	8	voter-education information, factual news report, accurate description of IEC procedure, opinion framed as opinion, correct debunk of a rumour

Gemini Batch API, with Google Search grounding enabled, working through the prompts in their seeded order. Labels come from the generation itself rather than from human annotation.

6.2.2 Translation and Quality Control

Rows were assigned to languages in rotation (isiZulu, Sepedi, English), so each statement appears in exactly one language and the three languages are roughly equal in size. English rows were kept as generated, Gemini 3.6 Flash translated the others, instructed to preserve meaning, tone and register, to copy proper nouns, numbers, dates and hashtags unchanged, and to add or remove nothing. Rows that failed translation (978) were excluded. Every model response was parsed as JSON, rows that failed parsing were isolated rather than repaired, and each row received an identifier derived from its topic, statement and prompt, so that statements from the same prompt can be grouped. No English source sample occurs more than once. A random sample of 50 statements, stratified by language and label (25 Sepedi and 25

isiZulu, seed 559243982), was back-translated into English and reviewed by people, confirming that the translations preserved the meaning of the original statements.

6.2.3 Final Corpus

The final corpus contains 13,997 statements, balanced across languages and labels (Table 4).

Table 4: Final corpus by language and label.

Language	Misinformation	Non-misinformation	Total
English	2,360	2,310	4,670
Sepedi	2,360	2,307	4,667
isiZulu	2,341	2,319	4,660
Total	7,061	6,936	13,997

6.3 Misinformation Detection Model

The detection component is a fine-tuned transformer classifier that labels each statement as misinformation (the positive class) or non-misinformation, so that fact-checkers can prioritise their review. The fine-tuning study settled two choices: which model to deploy, and whether to fine-tune it fully or to train a lightweight adapter on a shared base model. It compared three classical TF-IDF baselines, a South African decoder-only language model (MzansiLM-125M) and two African-language encoders (AfroXLMR-base-76L and AfroXLMR-large-76L), all trained and tested on the same split. Each transformer was fine-tuned in two ways, updating all of its parameters (full fine-tuning) and training only low-rank adapters (LoRA), each with three random seeds.

6.3.1 Data Preparation

Each statement was cleaned by removing emojis, replacing runs of exclamation marks with a full stop, removing all punctuation except full stops, collapsing repeated whitespace and converting to lower case. Statements shorter than three characters after cleaning (100) and 20 degenerate statements, in which one word is repeated fifteen times or more, were removed,

leaving 13,877: 7,008 misinformation and 6,869 non-misinformation, with 4,661 in English, 4,642 in Sepedi and 4,574 in isiZulu.

The statements were split 80/10/10 into training, validation and test sets, stratified by label and language (Table 5). Within each label–language group, statements kept their corpus order, which is grouped by prompt, and were cut into consecutive segments that were then shuffled, so statements from the same prompt stay together: only 22 of the 1,408 prompts contribute to more than one split, which limits leakage between near-identical statements. The validation set was used for checkpoint selection and early stopping, and the test set only for the final evaluation.

Table 5: Training, validation and test sets.

Set	**Total**	**Non-misinfo.**	**Misinfo.**	**English**	**Sepedi**	**isiZulu**
Training	11,099	5,494	5,605	3,728	3,713	3,658
Validation	1,385	686	699	465	463	457
Test	1,393	689	704	468	466	459

6.3.2 Models

Classical baselines. Logistic regression and a linear support vector machine (SVM), both with balanced class weights, and multinomial naive Bayes were trained on TF-IDF features of word unigrams and bigrams, using terms that occur in at least two training statements and sublinear term-frequency scaling. For the SVM, a softmax over its decision score gave the probabilities needed for ROC AUC. Classifiers of this kind are the usual starting point in South African misinformation and hate-speech studies [15, 16], so they show how much the transformers add.

Transformers. Three transformers were compared (Table 6). MzansiLM-125M is a decoder-only model trained from scratch on all eleven official South African languages [67], the only model in the study built specifically for them. AfroXLMR-base-76L and AfroXLMR-large-76L (Hugging Face checkpoints afro-xlmr-base-76L_script and afro-xlmr-large-76L) adapt the multilingual encoder XLM-R to 76 languages widely spoken in Africa, including Sepedi and isiZulu, through continued masked-language-model training [68]. AfroXLMR outperformed other multilingual models and zero-shot GPT-4 on sentiment classification in South African languages [30], further adaptation to social media text improves it [69], and fine-tuned encoders generally outperform decoder-only LLMs on classification [27], so the encoders were expected to lead.

Table 6: Transformer models.

Model	Architecture	Pre-training languages
MzansiLM-125M	Decoder-only, 125M parameters	All 11 official South African languages
AfroXLMR-base-76L	Encoder (XLM-R base)	76 languages widely spoken in Africa
AfroXLMR-large-76L	Encoder (XLM-R large)	76 languages widely spoken in Africa

6.3.3 Fine-Tuning

Each transformer received a sequence-classification head with two outputs. Inputs were truncated at 256 tokens, which affects only 46 of the 13,877 statements (0.3%), and batches were padded to their longest sequence. Training used the AdamW optimiser with weight decay 0.01, a linear learning-rate schedule with warm-up and 16-bit mixed precision, for up to ten epochs. The checkpoint with the highest validation F1-score for the misinformation class was kept, and training stopped early when that score had not improved for two consecutive epochs. The loss was cross-entropy weighted by class,

$$w_c = \frac{N}{K n_c} \tag{1}$$

where N is the number of training statements, $K = 2$ is the number of classes and n_c is the number of training statements in class c. Because the classes are nearly balanced, both weights are close to one. Table 7 gives the settings that differ between models. AfroXLMR-large used a smaller per-device batch with gradient accumulation and gradient checkpointing to fit in GPU memory, keeping an effective batch size of 16.

Table 7: Fine-tuning settings by model.

Model	Batch	Accumulation	Learning rate	Warm-up	Gradient checkpointing
MzansiLM-125M	16	1	2×10^{-5}	10%	No
AfroXLMR-base-76L	16	1	2×10^{-5}	10%	No
AfroXLMR-large-76L	8	2	1×10^{-5}	15%	Yes

Full fine-tuning and LoRA. Full fine-tuning updates every parameter of the model. Low-Rank Adaptation (LoRA) instead freezes the pre-trained weights and learns a low-rank update $\Delta W = BA$ for selected weight matrices,

which cuts the number of trainable parameters sharply [70]. LoRA was applied to the query and value projections of every attention layer, with rank $r = 64$, scaling factor $\alpha = 128$ (a scale of $\alpha/r = 2$) and dropout 0.1, and without training bias terms. The classification head remained fully trainable. Under this configuration LoRA trains between 1.00% and 2.54% of each model's parameters (Table 8). Both methods used the settings in Table 7. Setting $\alpha = 2r$ follows Shuttleworth et al., who found that it produced fewer intruder dimensions and better generalisation than a smaller α [54].

Table 8: Trainable parameters under LoRA ($r = 64$, $\alpha = 128$). Totals include the adapters and the classification head.

Model	**Trainable**	**Total**	**Trainable (%)**
MzansiLM-125M	3,257,344	128,266,752	2.54
AfroXLMR-base-76L	2,951,426	296,356,612	1.00
AfroXLMR-large-76L	7,343,106	567,235,588	1.29

LoRA and full fine-tuning can produce structurally different weight updates even when their accuracy is similar [54], while parameter-efficient methods lower the cost of training and deploying models when data and compute are limited [71], so the study measures the difference rather than assuming the two are equivalent. Every combination of transformer and method was trained with three seeds (619, 436 and 777), giving 18 fine-tuning runs, because fine-tuned AfroXLMR models vary considerably between runs in low-resource settings [72].

6.3.4 Evaluation and Model Selection

All models were evaluated once on the test set after training. A statement was predicted as misinformation when that class had the higher predicted probability, and the report gives accuracy, the precision, recall and F1-score of the misinformation class, and ROC AUC. F1-score is the primary measure and the one used to select checkpoints, because both errors matter for triage: a missed misinformation statement is likely to receive less attention in the review queue, while a false alarm uses up fact-checkers' time. Results are broken down by language, with Sepedi and isiZulu also grouped as African languages, and the three seeds are summarised by the mean and sample standard deviation of F1. To compare how the methods train, the analysis records the epoch chosen by early stopping and the training and validation

losses at that epoch, a validation loss well above the training loss indicates overfitting.

The architecture was chosen on mean F1-score over the three seeds, checking that Sepedi and isiZulu did not fall far behind English. The fine-tuning method was then chosen on expected behaviour on unseen data, weighing the gap between training and validation loss and the share of parameters trained against the loss in F1-score and the seed-to-seed spread, because the test set shares the synthetic origin of the training data, Section 7.1.6 reports the choice. The baselines were implemented with scikit-learn and the transformers with PyTorch, Hugging Face Transformers (version 4) and the PEFT library, trained on a GPU in Google Colab.

6.4 System Architecture

The prototype is a decoupled, multi-service architecture with four layers behind a single HTTPS edge: an application and ingestion layer, an off-chain storage layer, a monitoring layer and a permissioned blockchain layer (Figure 2), implementing Objective 4 (SRQ3). Ingestion, inference and ledger submission are decoupled through an Apache Kafka event-streaming pipeline [73], because election traffic surges unpredictably and synchronous submission to a blockchain gateway would cause backpressure and timeouts [74], Kafka's publish–subscribe model provides durability and rate smoothing between stages [75]. Each message is signed with a keyed hash (HMAC) by its producer, with a separate key for each hop, so a claim or label cannot be injected or altered on the queue without detection, and the submission service discards repeated message identifiers so that redelivered messages do not create duplicate reports.

6.4.1 Blockchain Layer

A single command brings up the Fabric network, with one certificate authority per organisation and CouchDB as the default world-state database, a private three-node IPFS swarm behind a storage service, an SDK sidecar that mediates all interaction with the ledger, a REST gateway that is the sole point of contact for stakeholder organisations and a web edge that is the only public entry point. The consortium size (3 to 30 organisations) is set at start-up, and every transaction must be endorsed by any two member organisations. The founding organisations register at start-up, after that, a new organisation is admitted only when two-thirds of the registered organisations vote for it (Section 6.6.2). In the single-host prototype one SDK

sidecar holds the signing identities of every organisation, whereas in a production consortium each organisation runs its own sidecar and keeps its keys on its own infrastructure (Section 8.2.1).

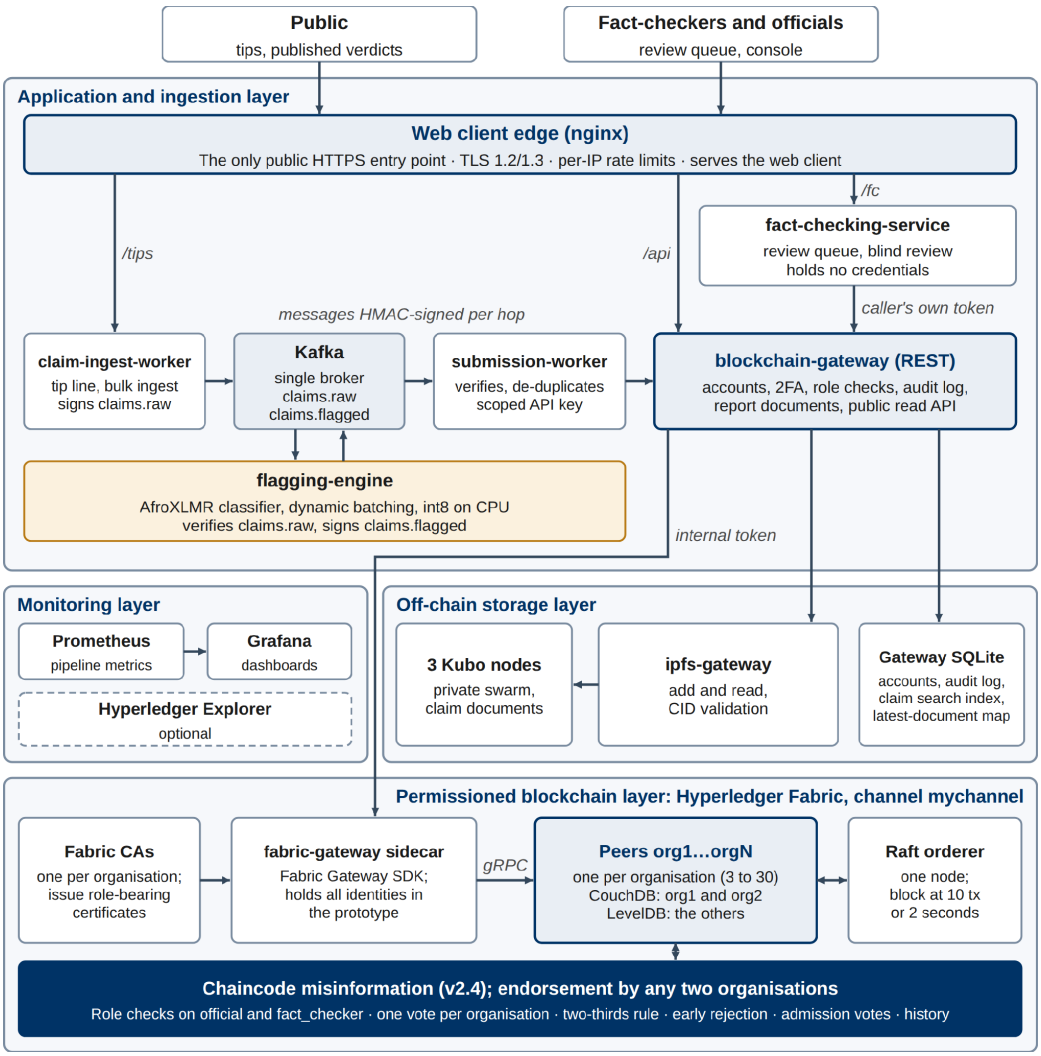


Figure 2: System architecture

Figure 2 has three trust boundaries. The web edge is the only public entry point, behind it, the blockchain gateway authenticates people and services and applies their roles and the chaincode checks the role attribute in the caller's certificate again on every endorsing peer, so the ledger does not have to trust the gateway. In the prototype these boundaries are logical, in a production consortium each organisation holds its own gateway, sidecar, peer and keys.

6.4.2 Application Pipeline

An ingestion service publishes claims to a message topic, taking anonymous tips from the public and bulk submissions authenticated with an API key. A detection service runs the classifier (Section 6.3) with dynamic request batching, and a submission service submits every labelled claim to the gateway as a report, whether flagged or not, so the label informs review without deciding which claims reach it. A fact-checking service gives each organisation a review queue sorted by confidence and applies blind review, hiding other organisations' votes until the reviewer's organisation has voted or the report has closed, every vote must carry written reasoning. A web client serves fact-checkers and officials, together with a public portal where anyone can submit a tip anonymously and read closed reports with each organisation's vote, reasoning and evidence, the transaction history and an integrity check. Claims under review are not published, and reviewers appear only as their organisation.

6.5 Hyperledger Fabric Blockchain Framework

The persistence layer uses Hyperledger Fabric [76], a permissioned consortium blockchain framework. Two design decisions, addressing SRQ2 and Objective 3, are justified below: choosing a permissioned rather than a permissionless blockchain such as Bitcoin or Ethereum (Section 6.5.1), and choosing Hyperledger Fabric over the other permissioned frameworks (Section 6.5.2).

6.5.1 Permissioned Blockchain Justifications

No transaction fees. Permissionless blockchains such as Bitcoin [77] and Ethereum [78] impose variable, unpredictable transaction fees, called gas fees in proof-of-stake systems [79] such as Ethereum and miner fees in proof-of-work systems such as Bitcoin, which make continuous election monitoring financially unviable for non-profit fact-checkers and public institutions.

Institutional identity and non-repudiation. Election integrity requires strict institutional accountability. Fabric's permissioned Membership Service Provider (MSP) issues X.509 digital certificates to verified organisations, so every submission and vote is tied to a verified entity rather than an anonymous public key [76], [49].

Throughput. The system must keep pace with the velocity at which social media data arrives [80]. Permissionless consensus introduces probabilistic finality and high confirmation latency, whereas permissioned frameworks allow specialised ordering services with sub-second block confirmation,

deterministic finality and no risk of chain forks [81], [82].

6.5.2 Hyperledger Fabric Justification

Hyperledger Fabric, R3 Corda, MultiChain and Quorum are the leading enterprise permissioned frameworks, and a systematic comparison found the choice between them largely use-case specific [83]. For this system the deciding requirement is that role checks are enforced in chaincode. In a consortium the gateway is not a trusted boundary: an organisation could modify its own gateway to skip role checks, or bypass it by submitting transactions directly to a peer. In Fabric's execute–order–validate model, every endorsing peer independently evaluates the role attribute in the caller's X.509 certificate through the Client Identity chaincode library [76], [84], and a transaction from a caller without the required role fails endorsement and is never committed, whichever client submitted it. This guarantee rests on each organisation controlling its own signing keys.

The alternatives do not provide this. Corda's point-to-point model shares a transaction only with its counterparties [85], which rules out the network-wide record on which every member votes and from which the public portal reads. Quorum's smart-contract permissioning governs who may join the network and transact [86], and MultiChain grants global permissions to addresses [87], neither evaluates certificate role attributes inside application logic.

6.6 Chaincode and On-Chain Data Model

Raw claim text is never stored on the blockchain. The claim, the full inference record, fact-checkers' reasoning and supporting evidence reside off-chain in an InterPlanetary File System (IPFS) Kubo swarm [88], and only structured metadata and SHA-256 hashes are anchored on-chain, as in similar projects [49]. Storing large documents on-chain multiplies storage across all full nodes and creates scalability bottlenecks [50], [39], and ledger immutability conflicts with POPIA's minimality and retention requirements [44] (Section 2.5), keeping claim text off the ledger aligns the design with POPIA at the architectural level (Section 7.3). The design addresses Objectives 3, 5 and 6.

6.6.1 Report Identity and Off-Chain Storage

Every report is keyed by the content identifier (CID) of the first version of its off-chain document, which holds the claim and its AI inference metadata.

Each vote publishes a new version that adds the organisation's reasoning, evidence and a pseudonymous reviewer reference, the ledger key stays the first CID, and the gateway's database maps it to the latest version. The on-chain record carries a SHA-256 hash of the document's immutable core (the claim and the AI inference) and structured metadata: the AI label, confidence and model version, a submission timestamp, the submitting organisation, a 72-hour deadline, a status and the list of votes (organisation, outcome and transaction identifier). The verification endpoint recomputes the hash of the retrieved core and checks it against the ledger, so tampering with the claim or the AI's label is detected without exposing personal data on the immutable ledger [89].

These mechanisms provide four properties. **Integrity:** the hash check proves that the retrieved claim and inference are exactly what was anchored, while the consortium's vote establishes whether the claim is accurate. **Availability:** each document is added to one Kubo node and fetched by the other nodes on demand. **Confidentiality:** the swarm is private, its interfaces are bound to the local host, and access to documents is controlled at the gateway. **Privacy:** personal information stays in off-chain documents, which can be removed without touching the ledger (Section 7.3).

6.6.2 Consensus and Finalisation Paths

The ordering service uses Raft, a crash fault tolerant (CFT) protocol in which a leader–follower model among ordering nodes establishes a total order of transactions [90]. Raft is Fabric's recommended ordering service, with deterministic finality and no chain forks [91]. It was preferred over Byzantine fault tolerant (BFT) protocols because consortium members are known and vetted, so the threat model is crash failures rather than arbitrary malicious behaviour [92], and because Raft achieved lower latency and higher throughput than PBFT in Fabric [93], [94]. Raft does not tolerate Byzantine faults [95], but MSP-issued X.509 identities tie every transaction to a verified entity [76]. The orderer cuts a block at 10 transactions or 2 seconds, whichever comes first, the prototype runs one orderer, and a production consortium runs several hosted by different organisations.

Quorum threshold. Each registered organisation, including the one that submitted the report, may cast one vote on it: 1 if the evidence shows the claim is false or misleading, and 0 if it is accurate or not shown to be false. A claim is confirmed as misinformation when $\lceil 2N/3 \rceil$ of the N registered organisations vote 1, that is 2, 4, 7, 10 and 14 votes at 3, 5, 10, 15 and 20 organisations. A two-thirds threshold was chosen over a simple majority because a verdict of misinformation is a public statement on behalf

of the consortium and should carry broad agreement. It also guarantees that, with two or more members, the submitting organisation can never decide alone. The threshold uses the number of organisations registered at the time of each vote, and within an organisation the first official or fact-checker to vote casts its single vote. The same threshold governs the admission of new organisations.

Finalisation paths. A report opens as PENDING with a 72-hour review window and becomes UNDER_REVIEW at its first vote. After every vote, it becomes FINAL as misinformation as soon as the votes for misinformation reach the threshold (early acceptance), or REJECTED as soon as the threshold can no longer be reached (early rejection), which the public portal shows as not confirmed rather than as accurate. After the window, an official may close a report that is still open, by simple majority of the votes cast (at least two votes and no tie) or by marking it EXPIRED, and the gateway expires reports still PENDING in an hourly pass. The two-thirds rule is the primary path to a verdict, and the post-window route is a fallback that keeps reports from stalling. Table 9 summarises the paths.

Table 9: Report status transitions and finalisation paths. N is the number of registered organisations at the time of the vote, and the threshold is $\lceil 2N/3 \rceil$.

Path	**Trigger**	**Condition**	**Result**
Submission	A member organisation submits a report	New report identifier	PENDING, with a 72-hour deadline
First vote	A registered organisation votes	Report is PENDING	UNDER_REVIEW
Early acceptance	Any vote while the report is open, within or after the window	Votes of 1 reach the threshold	FINAL, label misinformation
Early rejection	Any vote while the report is open, within or after the window	Votes of 1 plus organisations yet to vote fall below the threshold	REJECTED: not confirmed, and not a finding of accuracy
Manual finalisation	An official, after the window, on a report still open	At least two votes and no tie, may be below the threshold	FINAL, with the label of the majority of votes cast
Expiry	An official, after the window, on a report still open or the gateway's hourly pass for reports still PENDING	None	EXPIRED: no verdict

Boundary cases. The chaincode and its unit tests (Section 6.8) settle the following. There is no abstain vote such that an organisation that does not vote leaves the report open until another path applies, and it still counts towards N. After the window, a report with fewer than two votes or a tie can only expire. Votes are accepted after the window while a report stays open, a second vote by the same organisation is refused, and a vote on a report that is FINAL, REJECTED or EXPIRED is refused. Because N is read at each vote, admitting a member raises the threshold for reports still open. The chaincode has no function for removing a member or revoking its access (Section 8.2.3).

6.7 Role-Based Access Control (RBAC)

Privileges are enforced at two levels. On the ledger, each organisation's certificate authority registers an official identity and a fact-checker identity whose X.509 certificates carry the role as an attribute, which the chaincode verifies before permitting report, vote and governance actions [76] (Section 6.5.2). At the gateway, people sign in with individual accounts (with optional two-factor authentication) and services use API keys, each bound to one organisation and role, and sign-ins, denied requests, votes, governance actions and account changes are written to a hash-chained audit log. Four roles are defined, of which the first two also exist on the ledger (SRQ2):

(i) Official. May submit reports and cast its organisation's vote, finalise or expire reports whose 72-hour window has passed, vote on and finalise the admission of organisations, and configure the initial founding organisation limit.

(ii) Fact-checker. May submit reports and cast its organisation's vote, once per report.

(iii) Observer. Has read-only access through the gateway, which answers its queries with the organisation's fact-checker identity and never invokes a transaction for it. Observer accounts are created by an official and support audit within the consortium, while the public portal provides scrutiny from outside it.

(iv) AI submitter. An API-key role for the submission service, which may submit reports and nothing else and signs with its organisation's fact-checker identity.

6.8 Testing and Verification

The contract logic is verified by 39 unit tests against a mock ledger state, covering registration, admission and report workflows, the automatic and manual decision paths, the on-chain history, the RBAC permissions and boundary cases such as the minimum quorum at each benchmarked consortium size, late votes and post-window finalisation below two-thirds. A further 81 gateway integration tests exercise the REST interface from the perspective of stakeholder organisations, and 26 tests cover the fact-checking service, the pipeline services, the storage service and the SDK sidecar. All 146 tests pass on the final code (Objective 6).

Ledger performance (Objective 7, SRQ5) was measured with Hyperledger Caliper 0.7.1, which drives a workload at a controlled send rate and records throughput, latency and success rate [96]. Fabric was benchmarked directly because published vendor figures are rarely comparable across platforms [97]. Each run deployed a fresh network with the project's deployment script, without the application services, and Caliper ran two rounds with two workers: 500 report submissions at a fixed 25 transactions per second (TPS), signed by the first organisation's fact-checker identity and endorsed by two organisations, and 500 queries of all reports at a fixed 50 TPS against the resulting 500-report ledger. Each configuration was run four to five times at 3, 5, 10, 15 and 20 organisations, once with CouchDB and once with LevelDB as the world-state database. In the CouchDB configuration, the first two organisations' peers, including the peer Caliper queried, used CouchDB and the others LevelDB, because the onboarding scripts do not configure CouchDB.

Caliper invokes the chaincode directly, so its results do not depend on the detection model. A separate functional test sent statements from the held-out test set through the detection service and the REST gateway, had organisations vote on each report, and checked its final status and hash verification, demonstrating that the full path completes. All tests and benchmarks ran on the device in Table 10.

7 Evaluation and Discussion

7.1 AI-Enabled Misinformation Detection

This section discusses the results of the fine-tuning study described in Section 6.3. All figures are measured on the held-out test set of 1,393 synthetic, machine-translated statements, and the transformer results are summarised over the three seeds (619, 436 and 777). Table 11 compares all models, Ta-

Table 10: Benchmark device specifications.

Specification	Value
CPU	AMD Ryzen 7 5800HS (8 cores / 16 threads)
RAM	10GB (+ 10GB swapfile)
Platform	WSL2 on Windows (Microsoft hypervisor)
OS	Linux 6.18.33.2-microsoft-standard-WSL2
Flagging engine device	CPU (int8 quantisation)
Ledger	Hyperledger Fabric test network (peers and orderer 2.5.16, contract v2.4), Fabric CA, one Raft orderer, CouchDB or LevelDB
Benchmark tool	Hyperledger Caliper 0.7.1, 2 workers

ble 12 gives the full results for AfroXLMR-large-76L, and Table 13 breaks its F1-score down by language.

Table 11: Test-set results by model and fine-tuning method. Transformer figures are means over three seeds, with the sample standard deviation of F1, the classical baselines are deterministic.

Model	Method	Accuracy	F1	AUC
TF-IDF + logistic regression	Classical	0.799	0.807	0.886
TF-IDF + linear SVM	Classical	0.814	0.822	0.899
TF-IDF + multinomial NB	Classical	0.816	0.827	0.898
MzansiLM-125M	Full	0.842	0.851 ± 0.003	0.918
MzansiLM-125M	LoRA	0.773	0.788 ± 0.003	0.863
AfroXLMR-base-76L	Full	0.851	0.859 ± 0.005	0.935
AfroXLMR-base-76L	LoRA	0.815	0.826 ± 0.010	0.901
AfroXLMR-large-76L	Full	0.895	0.899 ± 0.009	0.961
AfroXLMR-large-76L	LoRA	0.877	0.882 ± 0.015	0.940

7.1.1 Overall Performance

AfroXLMR-large-76L achieved the highest F1-score under both methods: 0.899 ± 0.009 with full fine-tuning and 0.882 ± 0.015 with LoRA, with a best single run of 0.906 (full fine-tuning, seed 777, AUC 0.963). Every AfroXLMR configuration outperformed the TF-IDF baselines (F1 0.807–0.827) except AfroXLMR-base with LoRA (0.826), which was level with the strongest

baseline, naive Bayes. This reverses the South African studies in Section 3.1, where classical classifiers performed best and the one transformer, trained on 1,500 translated sentences, reached at most 79% accuracy [15]. The larger election corpus (11,099 training statements) is the most likely reason. The ranking matches earlier findings that fine-tuned encoders, Afrocentric ones in particular, outperform decoder-only models on classification [30], [27], [52]: MzansiLM-125M came close to AfroXLMR-base under full fine-tuning (0.851 against 0.859) but fell below all three baselines under LoRA (0.788).

7.1.2 Full Fine-Tuning Versus LoRA

Full fine-tuning achieved the higher F1-score in all nine comparisons, but the margin narrowed as the models grew: 0.063 for MzansiLM-125M, 0.033 for AfroXLMR-base and 0.018 for AfroXLMR-large, on which LoRA recovered about 98% of the fully fine-tuned F1-score while training 1.29% of the parameters (Table 8). LoRA was less consistent across seeds, with a higher F1 standard deviation on both AfroXLMR models (0.010 against 0.005, and 0.015 against 0.009). On AfroXLMR-large the single-seed gap ranged from 0.002 (seed 619) to 0.038 (seed 436), and LoRA's own spread (0.865–0.893) exceeded the mean gap, so any single seed would have misstated the difference, which confirms the value of training with several seeds [54], [72].

Table 12: Test-set results for AfroXLMR-large-76L by fine-tuning method and seed.

Method	Seed	Accuracy	Precision	Recall	F1	AUC
Full	619	0.881	0.841	0.943	0.889	0.956
Full	436	0.901	0.890	0.918	0.903	0.963
Full	777	0.902	0.875	0.939	0.906	0.963
Full	Mean	0.895	0.869	0.933	0.899	0.961
LoRA	619	0.885	0.885	0.888	0.887	0.946
LoRA	436	0.858	0.833	0.899	0.865	0.929
LoRA	777	0.889	0.869	0.918	0.893	0.945
LoRA	Mean	0.877	0.862	0.902	0.882	0.940

7.1.3 Training Behaviour

The two methods also trained differently. Under full fine-tuning, the epoch selected by early stopping came early (on average epoch 3.0 to 5.0, depending on the model), and by then the validation loss was already well above the

training loss, by up to 0.96 on MzansiLM-125M, by 0.36–0.58 on AfroXLMR-base and by 0.40–0.49 on two of the three AfroXLMR-large runs, a sign of overfitting. The third AfroXLMR-large run (seed 619) was the exception, stopping at epoch 2 with no gap. LoRA runs peaked later (on average epoch 5.3 to 7.0) and stopped with training and validation losses close together (differences between −0.04 and +0.13). LoRA thus gives up a small amount of F1 for a much smaller gap between training and validation loss, although its results vary more between seeds.

7.1.4 Performance by Language

AfroXLMR-large-76L performed best on English and almost equally on Sepedi and isiZulu (Table 13). The gap of about 0.04 F1 between English and the African languages is modest and meets the selection criterion in Section 6.3. Unlike the sentiment results of Mabokela et al., in which AfroXLMR favoured the Nguni languages [31], Sepedi and isiZulu scored almost identically here, consistent with the 76-language model including both in pretraining. LoRA cost slightly more on the African languages (0.020) than on English (0.016). Each language contains different statements, and 49 of the 459 isiZulu and 13 of the 466 Sepedi test statements are unchanged English, so a parallel subset with the same claims in all three languages would isolate the pure language effect (Section 8.2.2).

Table 13: F1-score of AfroXLMR-large-76L by language, mean over three seeds.

Method	**English**	**Sepedi**	**isiZulu**	**Sepedi and isiZulu**
Full	0.926	0.885	0.889	0.887
LoRA	0.910	0.865	0.870	0.867
Difference (Full − LoRA)	0.016	0.020	0.019	0.020

7.1.5 Errors and Triage

For triage, a missed misinformation statement is labelled as not flagged and is likely to receive less attention in the review queue, while a false alarm costs fact-checkers' time (Section 6.3). Full fine-tuning's advantage on AfroXLMR-large came mainly from recall (mean 0.933 against 0.902 for LoRA), with similar precision (0.869 against 0.862). The confusion matrices show the same pattern: of the 704 misinformation statements in the test set, full

fine-tuning missed between 40 and 58 per seed and LoRA between 58 and 79, a mean of 47 against 69, with about 100 false alarms per run under both methods. The exception was seed 619, on which the two methods were nearly tied: full fine-tuning raised more false alarms there (126 against 81).

7.1.6 Choice of Model for the Framework

The framework selects AfroXLMR-large-76L with LoRA and retains the run from seed 777, which had the highest validation F1-score of the three LoRA runs (0.891, against 0.884 for seed 436). On the test set it reached an F1-score of 0.893 (precision 0.869, recall 0.918, AUC 0.945), with 0.920 on English, 0.881 on Sepedi and 0.878 on isiZulu, missing 58 of the 704 misinformation statements and raising 97 false alarms on the 689 accurate ones. Against full fine-tuning, LoRA gives up 0.018 in mean F1-score (0.882 against 0.899), mainly through recall (0.902 against 0.933), and varies more between seeds (standard deviation 0.015 against 0.009). The cost is contained because every screened claim reaches the review queue with its label and can still be reviewed and outvoted.

The choice rests on expected behaviour on real election posts rather than on the test score. Two of the three fully fine-tuned runs stopped with the validation loss 0.40–0.49 above the training loss, whereas all three LoRA runs stopped with the two losses within 0.04 of each other. Because the test set comes from the same synthetic pipeline as the training data, a model that has begun to fit that data can still score well on it. The LoRA model, which trains only 1.29% of the parameters and keeps the pre-trained weights frozen, is better placed to generalise to real posts, and its small adapter is cheap to retrain as real data becomes available [71]. Two counterpoints were weighed: the third full run (seed 619) stopped at epoch 2 with no gap, so overfitting is not inherent in full fine-tuning and LoRA can introduce intruder dimensions that cause forgetting, which setting $\alpha = 2r$ reduces [54]. Because the model supplies only a label, a confidence score and a version string, it is a replaceable component (principle P5, Section 5.1). The end-to-end test in Section 7.2.2 therefore used the fully fine-tuned checkpoint that the detection service held during development, and its ledger results do not depend on that choice, the serving check in Section 7.1.7 was run on the same checkpoint. The selected model is deployed by merging its adapter into the base model and updating the version string.

7.1.7 Limitations

The scores are measured on synthetic statements (Section 6.2), so evaluation on real election posts, including whether LoRA's advantage in training behaviour carries over, is the next step (Section 8.2.2). Because every statement comes from one generation process, the classifier may also draw on cues specific to the generating model, which evaluation on real posts will separate from the substance of claims.

The detection service runs the model int8-quantised on a CPU and scores claims as received, whereas the figures above come from floating-point models on a GPU and cleaned text. Scored on the same 1,393 test statements, the running service, which holds a fully fine-tuned checkpoint, reached an F1-score of 0.895 on cleaned text (0.921 on English, 0.888 on Sepedi and 0.875 on isiZulu) and 0.899 on raw text, with no significant difference between the two (McNemar's test, $\chi^2 = 0.3$). Both lie within the range of the fully fine-tuned runs (0.889–0.906), so int8 serving on raw input preserves the performance measured on the GPU.

7.2 Blockchain Layer

7.2.1 Ledger Performance under Hyperledger Caliper

Tables 14 and 15 give the Caliper results for the write and read rounds described in Section 6.8, as means and sample standard deviations over repeated runs. A confirmation run after a final contract change, which does not touch the benchmarked paths, had no failed transactions.

Writes. Write throughput stayed between 24.3 and 25.0 TPS at every consortium size, matching the 25 TPS send rate, and only 19 of 24,000 write transactions failed (0.08%). Average write latency rose only modestly with consortium size, from 0.32 to 0.42 s with CouchDB and from 0.24 s to at most 0.32 s with LevelDB. This path is independent of membership by design: the endorsement policy always requires two organisations, and every write carries the same fixed payload because claim text stays off-chain (Section 6.6). Average latency stayed well below the orderer's two-second batch timeout, consistent with blocks filling to their ten-transaction limit at 25 TPS, the slowest individual writes took up to 4.0 s. Voting, whose cost grows with the number of votes needed (14 at 20 organisations), is the next workload to benchmark (Section 8.2.4).

Reads. With LevelDB, the full-ledger query kept pace with the 50 TPS send rate at average latencies of 0.02–0.09 s. With CouchDB, throughput fell to 29.8–33.3 TPS and average latency rose to 4.20–5.83 s, because LevelDB is

Table 14: Caliper write round: 500 report submissions at a fixed 25 TPS send rate (achieved send rate 24.7–25.1 TPS). Mean ± standard deviation over runs, Failed is the mean number of failed transactions per run.

Orgs	State DB	Runs	Failed	Throughput (TPS)	Avg latency (s)	Max latency (s)
3	CouchDB	5	0.0	24.6 ± 1.0	0.32 ± 0.03	2.65 ± 1.25
3	LevelDB	5	0.6	25.0 ± 0.0	0.24 ± 0.02	0.99 ± 1.30
5	CouchDB	5	1.0	25.0 ± 0.0	0.34 ± 0.05	2.80 ± 1.32
5	LevelDB	5	0.6	24.5 ± 0.8	0.28 ± 0.03	2.71 ± 1.28
10	CouchDB	5	0.4	24.8 ± 0.3	0.34 ± 0.02	3.19 ± 0.07
10	LevelDB	5	1.0	24.9 ± 0.3	0.30 ± 0.07	2.22 ± 1.63
15	CouchDB	5	0.2	24.3 ± 1.0	0.40 ± 0.10	2.82 ± 1.21
15	LevelDB	5	0.0	24.5 ± 1.0	0.32 ± 0.05	2.36 ± 1.64
20	CouchDB	4	0.0	24.8 ± 0.3	0.42 ± 0.11	2.85 ± 1.29
20	LevelDB	4	0.0	24.9 ± 0.1	0.29 ± 0.03	1.20 ± 1.35

Table 15: Caliper read round: 500 queries of all reports at a fixed 50 TPS send rate. Mean ± standard deviation over runs, Failed is the mean number of failed queries per run.

Orgs	State DB	Runs	Failed	Throughput (TPS)	Avg latency (s)	Max latency (s)
3	CouchDB	5	0.0	32.4 ± 4.9	4.72 ± 2.08	7.50 ± 3.35
3	LevelDB	5	0.0	50.1 ± 0.0	0.02 ± 0.00	0.02 ± 0.00
5	CouchDB	5	0.0	32.9 ± 3.8	4.20 ± 0.85	6.88 ± 1.79
5	LevelDB	5	0.0	50.1 ± 0.0	0.09 ± 0.11	0.31 ± 0.50
10	CouchDB	5	0.0	31.9 ± 3.5	4.66 ± 1.34	7.56 ± 2.18
10	LevelDB	5	0.0	48.6 ± 3.3	0.03 ± 0.03	0.82 ± 1.78
15	CouchDB	5	0.0	33.3 ± 1.4	4.64 ± 0.63	6.66 ± 0.72
15	LevelDB	5	0.0	50.1 ± 0.0	0.02 ± 0.00	0.03 ± 0.00
20	CouchDB	4	0.0	29.8 ± 3.8	5.83 ± 1.26	8.71 ± 2.19
20	LevelDB	4	0.0	47.9 ± 4.5	0.02 ± 0.00	0.05 ± 0.04

embedded in the peer whereas CouchDB is a separate database reached over HTTP that returns each record as a JSON document. No read queries failed. Each query returns every report, so its cost depends on the size of the ledger (500 reports in every run) rather than on the number of organisations, and

read latency showed no trend with consortium size. The gateway caches the result for five seconds on the public read path, and paginated, status-based queries bound the cost of a read in production (Section 8.2.3).

7.2.2 End-to-End Verification

To show that the components work together, 100 statements from the held-out test set were sent through the integrated system at three, five, ten, fifteen, twenty organisations consortium sizes: the detection service, holding the fully fine-tuned checkpoint, labelled each statement, the first organisation submitted it through the REST gateway, which stored the document in IPFS and recorded the report on Fabric, and the organisations voted. A smoke test also sent one public tip through the full asynchronous path (ingestion, Kafka, detection, submission, gateway, IPFS and Fabric).

All 100 reports were accepted on-chain and reached FINAL status after $\frac{2}{3}$ majority votes in total were reached. Within the review window FINAL can be reached only through early acceptance, so this test exercised that path, the unit tests cover early rejection and the post-window paths (Section 6.8). The model's label matched the ground-truth label for all six statements. The votes were scripted, Section 7.1 reports the model's accuracy.

7.3 POPIA Alignment

Four design choices align the architecture with conditions of the Protection of Personal Information Act (POPIA), and Section 7.3.2 sets out the legal questions that precede a real deployment.

7.3.1 Design Choices

Minimality (section 10). Only a SHA-256 content hash, a content identifier (CID), the AI label, confidence and model version, timestamps, the report's status and deadline, and each organisation's MSP identifier and vote are written on-chain, the claim text and evidence live off-chain in IPFS.

Purpose specification (section 13). The on-chain schema is fixed by the contract, with no free-form field for claim text or personal details, the only free-text fields, an admission request's organisation name and type and the model version string do not describe individuals.

Retention (section 14). Because documents are off-chain, their retention is governed by the operators of the IPFS nodes rather than by the chaincode. Each vote adds a new document version while older versions remain retrievable, and the gateway's search index holds a further copy of

the claim text, so a deletion is applied on every node and on the index. Section 8.2.3 automates this with a retention schedule.

Integrity and confidentiality (section 19). Content hashing and content addressing give a two-layer tamper check for the claim and the AI inference (Section 6.6.1). The Fabric channel and the public web edge use TLS, internal service traffic, including Kafka and the SDK sidecar, is authenticated rather than encrypted, by HMAC signatures and a shared token; and certificate-based roles in the chaincode restrict who may submit, vote, finalise or govern.

7.3.2 Open Questions

Several points raised in Section 2.5 are matters for a legal assessment ahead of real deployment. Claims may name individuals or allege offences, which POPIA treats as special personal information (section 26). The tip form asks people not to include personal details, and because the public portal publishes closed reports as submitted, a redaction step before publication is planned (Section 8.2.4). The lawful basis for processing (section 11), the authorisation for special personal information (section 27) and the reach of the journalistic exclusion (section 7) to a shared monitoring infrastructure are legal determinations rather than design ones. Correction and deletion requests (section 24) are served off-chain through the retention mechanism in Section 8.2.3, and whether the CID and hash that remain on the ledger count as personal information after a document is removed is the open question identified in Section 2.5.

8 Conclusion & Future Work

8.1 Conclusion

This study set out to determine how an integrated AI and blockchain framework could support transparent, tamper-evident and multilingual monitoring of election misinformation in South Africa. Its answer is the conceptual framework in Section 5, in which a model advises, independent organisations decide jointly and a permissioned ledger records, together with a prototype that instantiates it. On a synthetic election corpus in English, Sepedi and isiZulu, AfroXLMR-large-76L achieved the highest mean test F1-score (0.899 with full fine-tuning and 0.882 with LoRA, against 0.807–0.827 for the TF-IDF baselines) and serves as a triage tool whose label sits beside the fact-checkers' votes. The ledger anchors claims by hash with documents

held off-chain, enforces roles in chaincode and confirms misinformation only when two-thirds of members agree; report submission kept pace with a 25 TPS load up to 20 organisations, and six statements passed end to end from detection to a FINAL verdict.

8.1.1 Answers to the Research Questions

Section 4.2 asked how an integrated AI and blockchain framework can be designed and evaluated for tamper-resistant, transparent, multilingual monitoring of election misinformation. The answer is the separation in Section 5: a model that advises, organisations that decide jointly and a ledger that records. Table 16 sets the framework's requirements against the evidence, Table 17 records the status of each objective, and the sub-questions are answered as follows.

- **SRQ1.** A fine-tuned Afrocentric encoder suits this setting: AfroXLMR-large-76L led every comparison, and LLM-generated, machine-translated data addressed the lack of a labelled corpus.
- **SRQ2.** A permissioned Fabric consortium with certificate-based roles, any-two endorsement and off-chain documents balances integrity, privacy and accessibility, and report submission held its rate up to 20 organisations; voting load and multi-host deployment are the next stage of evaluation.
- **SRQ3.** The model can be integrated through a signed, asynchronous pipeline that records its label, confidence and version beside organisations' votes, which the functional test and the smoke test exercised end to end.
- **SRQ4.** Hashing, content addressing, off-chain storage and pseudonymous reviewer references keep claim text and individuals off the ledger while preserving an auditable record, aligning the architecture with POPIA; deletion, redaction and the legal basis are the remaining steps towards full compliance (Section 7.3.2).
- **SRQ5.** The unit and functional tests demonstrate transparency, traceability and auditability in the running prototype. By design, the framework improves on a centrally held record in auditability, since any member can check a report's hash and transaction history and no single custodian can rewrite them, and in institutional accountability, since each vote is tied on the ledger to an organisation's identity; this comparison is analytical rather than experimental (Table 17). Data quality

remains the responsibility of the reviewers, because a ledger preserves what it is given [17], and the framework's effect on public trust is a question for user evaluation (Section 8.2.2).

Table 16: Framework requirements set against the evidence.

Requirement (Section 4.5)	**Framework element**	**Evidence**	**Remaining limit**
R1 Multilingual processing with scarce data	AfroXLMR-large-76L with LoRA, trained on a synthetic corpus in three languages	Mean F1-score 0.882, retained run 0.893; Sepedi and isiZulu about 0.04 below English on different statements (Tables 11 to 13)	Synthetic data only; LoRA not yet served; code-switching and audio not covered
R2 Classification supports verification	Advisory label and confidence; votes by member organisations decide	Chaincode unit tests; functional test of six statements	Binary vote with no rating scale (Section 2.2)
R3 Shared record, no single custodian	Fabric consortium; any-two endorsement; two-thirds vote	Caliper up to 20 organisations: submission kept pace with 25 TPS (Tables 14 and 15)	Single host, one orderer and one signing service; one gateway maps reports to their latest documents; voting not load-tested
R4 Traceable decisions	On-chain votes and history; integrity check; public portal; blind review	Verification and public-interface tests; hash checks in the functional test	Reasoning and evidence not anchored; ledger identifies organisations, not individuals; no correction record (Section 2.5)
R5 Compliance with POPIA	On-chain and off-chain split; fixed schema; reviewer pseudonyms	Design analysis only (Section 7.3)	No deletion route or redaction; no legal assessment

8.1.2 Overall

The framework addresses the trust deficit and fragmentation in electoral monitoring through one separation of responsibilities: AI triages, people decide jointly, and every decision is recorded where no single member can alter

it. The ledger guarantees that a decision has not been altered, and the consortium vote carries responsibility for whether it is right; the hybrid on-chain/off-chain data model reconciles that permanence with POPIA at the architectural level. The prototype shows that the framework can be built from existing components and that report submission scales with the consortium. The next stage, set out in Section 8.2, takes it from a single-host prototype trained on synthetic data to a multi-host deployment evaluated on real election posts and with working fact-checkers, serving the selected LoRA model and anchoring fact-checkers' reasoning and evidence on the ledger.

8.2 Future Work

The following extensions build directly on the prototype:

8.2.1 Multi-Host and Inter-Institution Deployment

Deploy each organisation's peer, certificate authority and signing service on its own infrastructure, with several Raft orderers hosted by different organisations, and repeat the Caliper benchmarks under real inter-institution network conditions. This would also remove the single signing service that holds every organisation's keys in the prototype.

8.2.2 Real Election Data and User Evaluation

Evaluate the detector on real, consented election posts, including code-switched text and, eventually, transcribed audio, and evaluate the web client with working fact-checkers to establish whether the shared ledger changes how they collaborate and whether the public trusts its verdicts.

8.2.3 Identity, Revocation and Retention

The gateway already identifies individual users, but the ledger records only the organisation and the role. Future work should issue per-user Fabric identities with role attributes evaluated in chaincode, add certificate revocation to each organisation's membership service provider, automate an off-chain retention schedule with logged deletion, and replace the full-ledger query with paginated, status-based queries.

8.2.4 Record Integrity, Verdict Model and Voting Performance

Anchor a hash of each organisation's reasoning and evidence in its vote transaction, and record each report's latest document version on the ledger, so that the full basis of a verdict is tamper-evident and does not depend on one gateway's database. Add an appended correction record (Section 2.5) and consider a rating scale richer than a binary vote (Section 2.2). Set a minimum level of participation for the post-window route, add a redaction step before publication, and give observers a certificate of their own. Extend the Caliper workload with voting, to measure how the time to reach a verdict and the rate of commit conflicts grow with the size of the consortium.

Individual Contributions

Tumiso Mokautu: AI training, AI Research, Prototype Development
Josh Cloete: Blockchain Research, Blockchain Development, Blockchain Testing
Tshiamo Monametsi: Literature Review, AI and Blockchain Research

GitHub repository:
https://github.com/dsfsi/Blockchain_Enabled_Framework_for_Misinformation_Monitoring-cos700_26.git

References

[1] G. Gondwe, "AI-Driven Disinformation and Political Influence on WhatsApp in South Africa's 2024 Elections," *The International Journal of Press/Politics*, pp. 1–20, Nov. 2025. Published online. doi: 10.1177/19401612251395434.

[2] A. Monnier and C. Ségur, "Challenges for Fact-checking: Beyond False/True Verification," *InMedia*, vol. 10, 2025. doi: 10.4000/145d4.

[3] N. Newman, A. Ross Arguedas, C. T. Robertson, R. K. Nielsen, and R. Fletcher, "Reuters Institute Digital News Report 2025." Reuters Institute for the Study of Journalism, University of Oxford, Oxford, U.K., June 2025. doi: 10.60625/risj-8qqf-jt36.

[4] M. Small, "Peace and Security Issues in South Africa's 2024 National and Provincial Elections: Security Planning, Implementation, and Re-

sults," *Journal of African Elections*, vol. 23, no. 2, pp. 47–79, Dec. 2024. doi: 10.20940/JAE/2024/v23i2a3.

[5] W. Bird and T. Smith, "Media Coverage of South Africa's 2024 Elections: Setting Standards for a Media Performance Review," *Journal of African Elections*, vol. 23, no. 2, pp. 167–188, Dec. 2024. doi: 10.20940/JAE/2024/v23i2a8.

[6] E. Porter and T. J. Wood, "The global effectiveness of fact-checking: Evidence from simultaneous experiments in Argentina, Nigeria, South Africa, and the United Kingdom," *Proceedings of the National Academy of Sciences*, vol. 118, no. 37, Art. no. e2104235118, Sept. 2021. doi: 10.1073/pnas.2104235118.

[7] C. N. Tatchou, C. E. Uzuegbunam, C. R. Ononiwu, and E. M. Masaki, "Fact-Checkers' Perceptions of Their Roles and Challenges in Cameroon, Kenya, Nigeria, and South Africa," *Journalism Practice*, pp. 1–19, May 2026. Published online. doi: 10.1080/17512786.2026.2668491.

[8] D. M. Markowitz, T. R. Levine, K. B. Serota, and A. D. Moore, "Cross-checking journalistic fact-checkers: The role of sampling and scaling in interpreting false and misleading statements," *PLOS ONE*, vol. 18, no. 7, Art. no. e0289004, July 2023. doi: 10.1371/journal.pone.0289004.

[9] S. Lee, A. Xiong, H. Seo, and D. Lee, " 'Fact-checking' fact checkers: A data-driven approach," *Harvard Kennedy School (HKS) Misinformation Review*, vol. 4, no. 5, Oct. 2023. doi: 10.37016/mr-2020-126.

[10] P. B. Brandtzaeg and A. Følstad, "Trust and distrust in online fact-checking services," *Communications of the ACM*, vol. 60, no. 9, pp. 65–71, Sept. 2017. doi: 10.1145/3122803.

[11] C. Martiny, T. Rolfe, B. Zerie, A. Gallagher, and H. Schwertheim, "Online Gendered Abuse and Disinformation During the 2024 South African Elections." Institute for Strategic Dialogue (ISD), London, U.K., 2024.

[12] S. Rananga, B. Isong, A. Modupe, and V. Marivate, "Misinformation Detection: A Review for High and Low-Resource Languages," *Journal of Information Systems and Informatics*, vol. 6, no. 4, pp. 2892–2922, Dec. 2024. doi: 10.51519/journalisi.v6i4.931.

[13] M. Malange, S. Rananga, M. S. Mbooi, B. Isong, and V. Marivate, "Investigating the Effectiveness of Detecting Misinformation on Social

Media using Tshivenda Language," in *2024 IST-Africa Conference (IST-Africa)*, pp. 1–5, 2024. doi: 10.23919/IST-Africa63983.2024.10569873.

[14] L. Mokoena, S. Rananga, and M. Mbooi, "Misinformation Detection in COVID-19 News for Low-Resource Languages: A Sesotho Case Study," in *2025 IST-Africa Conference (IST-Africa)*, pp. 1–8, 2025. doi: 10.23919/IST-Africa67297.2025.11060466.

[15] M. Manaswe, S. Rananga, and T. Sindane, "Detecting COVID-19 Misinformation in Sepedi Short Texts Using Data Augmentation Techniques," in *2025 IST-Africa Conference (IST-Africa)*, pp. 1–7, 2025. doi: 10.23919/IST-Africa67297.2025.11060465.

[16] M. R. Sethosa, S. Rananga, and M. S. Mbooi, "Hate Speech Detection in isiZulu: A Semantic Approach," in *2025 IST-Africa Conference (IST-Africa)*, pp. 1–9, 2025. doi: 10.23919/IST-Africa67297.2025.11060561.

[17] F. Lumineau, W. Wang, and O. Schilke, "Blockchain Governance—A New Way of Organizing Collaborations?," *Organization Science*, vol. 32, no. 2, pp. 500–521, Mar.–Apr. 2021. doi: 10.1287/orsc.2020.1379.

[18] J. Botha, "Disinformation and Democracy in Africa and South Africa," in *Disinformation, Misinformation, and Democracy: Legal Approaches in Comparative Context* (R. J. Krotoszynski, Jr., A. Koltay, and C. Garden, eds.), pp. 270–310, Cambridge, U.K.: Cambridge University Press, 2025. doi: 10.1017/9781009373272.017.

[19] K. Allen and J. le Roux, "Under the influence? Online mis/disinformation in South Africa's May 2024 election." Institute for Security Studies, Southern Africa Report 61, Dec. 2024.

[20] A. Mare, "Same voices, different contestants: Media and the 2024 elections in South Africa," *Journal of African Elections*, vol. 23, no. 2, pp. 80–105, 2024. doi: 10.20940/JAE/2024/v23i2a4.

[21] A. A. Hamed, M. Zachara-Szymanska, and X. Wu, "Safeguarding authenticity for mitigating the harms of generative AI: Issues, research agenda, and policies for detection, fact-checking, and ethical AI," *iScience*, vol. 27, Art. no. 108782, Feb. 2024.

[22] J. Kaplan, "More speech and fewer mistakes." Meta Newsroom, Jan. 7, 2025. [Online]. Available: https://about.fb.com/news/2025/01/meta-more-speech-fewer-mistakes/.

[23] O. Vinhas and M. Bastos, "The WEIRD governance of fact-checking and the politics of content moderation," *New Media & Society*, vol. 27, no. 5, pp. 2768–2787, 2025. doi: 10.1177/14614448231213942.

[24] P. Sahitaj, I. Maab, J. Yamagishi, J. Kolanowski, S. Möller, and V. Schmitt, "Towards automated fact-checking of real-world claims: Exploring task formulation and assessment with LLMs." arXiv:2502.08909, 2025.

[25] B. Wasike, "You've been fact-checked! Examining the effectiveness of social media fact-checking against the spread of misinformation," *Telematics and Informatics Reports*, vol. 11, Art. no. 100090, 2023.

[26] C. Jia, A. Y. Lee, R. C. Moore, C. H.-S. Decatur, S. X. Liu, and J. T. Hancock, "Collaboration, crowdsourcing, and misinformation," *PNAS Nexus*, vol. 3, no. 10, Art. no. pgae434, Oct. 2024. doi: 10.1093/pnasnexus/pgae434.

[27] S. Raza, D. Paulen-Patterson, and C. Ding, "Fake news detection: comparative evaluation of BERT-like models and large language models with generative AI-annotated data," *Knowledge and Information Systems*, vol. 67, no. 4, pp. 3267–3292, 2025. doi: 10.1007/s10115-024-02321-1.

[28] M. Vallayil, P. Nand, W. Q. Yan, and H. Allende-Cid, "Explainability of Automated Fact Verification Systems: A Comprehensive Review," *Applied Sciences*, vol. 13, no. 23, Art. no. 12608, Nov. 2023. doi: 10.3390/app132312608.

[29] R. Sultana and T. Nishino, "Identification of fake news in low-resource language: A deep learning approach," in *Proc. 15th Int. Congr. Advanced Applied Informatics Winter (IIAI-AAI-Winter)*, (Bali, Indonesia), pp. 322–329, 2023. doi: 10.1109/IIAI-AAI-WINTER61682.2023.00066.

[30] K. R. Mabokela, T. Celik, and M. Primus, "Sentiment Analysis in African Languages: Evaluating Generative AI and Afrocentric Language Models," in *South African Computer Science and Information Systems Research Trends (SAICSIT 2025)* (A. Gerber, ed.), Communications in Computer and Information Science, pp. 259–276, Cham, Switzerland: Springer, 2026. doi: 10.1007/978-3-031-96262-2_17.

[31] K. R. Mabokela, M. Primus, and T. Celik, "Advancing sentiment analysis for low-resourced African languages using pre-trained language mod-

els," *PLOS ONE*, vol. 20, no. 6, Art. no. e0325102, June 2025. doi: 10.1371/journal.pone.0325102.

[32] R. Nair, S. N. Zafrullah, P. Vinayasree, P. Singh, M. M. A. Zahra, T. Sharma, and F. Ahmadi, "Blockchain-based decentralized cloud solutions for data transfer," *Computational Intelligence and Neuroscience*, vol. 2022, Art. no. 8209854, 2022.

[33] X. Wang, H. Xie, S. Ji, L. Liu, and D. Huang, "Blockchain-based fake news traceability and verification mechanism," *Heliyon*, vol. 9, no. 7, Art. no. e17084, July 2023. doi: 10.1016/j.heliyon.2023.e17084.

[34] A. Mahajan *et al.*, "ChainSEAL: A blockchain-based secure and adaptive evidence ledger," in *Proc. 18th Int. Conf. COMmunication Systems and NETworks (COMSNETS), Poster Track*, 2026. doi: 10.1109/COMSNETS67989.2026.11417770.

[35] N. Sekamogeng and C. B. Egbude, "Votes, voices and views: A review of the management of the South African 2024 national and provincial elections," *Journal of African Elections*, vol. 23, no. 2, pp. 150–166, 2024. doi: 10.20940/JAE/2024/v23i2a7.

[36] B. Thukuse, "Balancing Freedom of Expression and Accountability: The Legitimacy of Penalising Media Organisations for Disseminating 'Fake News' in South Africa," *Journal of Public Administration and Development Alternatives*, vol. 10, no. 3, pp. 207–220, Dec. 2025. doi: 10.55190/JPADA.2025.454.

[37] F. Casino, T. K. Dasaklis, and C. Patsakis, "A systematic literature review of blockchain-based applications," *Telematics and Informatics*, vol. 36, pp. 55–81, 2019.

[38] K. Almiani, S. B. Mirza, N. Siyam, S. A. Al-Jaziri, O. Alqaryouti, and C. Zufferey, "Global adoption and impact of blockchain technology in government: Enhancing transparency, efficiency, and trust in public services." SSRN preprint 5719358 (not peer reviewed), 2025.

[39] S. Huckle and M. White, "Fake news: A technological approach to proving the origins of content, using blockchains," *Big Data*, vol. 5, no. 4, pp. 356–371, 2017. doi: 10.1089/big.2017.0071.

[40] A. Qayyum, J. Qadir, M. U. Janjua, and F. Sher, "Using blockchain to rein in the new post-truth world and check the spread of fake news," *IT Professional*, vol. 21, no. 4, pp. 16–24, 2019.

[41] D.-O. Jaquet-Chiffelle, E. Casey, and J. Bourquenoud, "Tamperproof timestamped provenance ledger using blockchain technology," *Forensic Science International: Digital Investigation*, vol. 33, Art. no. 300977, 2020. doi: 10.1016/j.fsidi.2020.300977.

[42] M. Hasnain, F. R. Albogamy, S. S. Alamri, I. Ghani, and B. Mehboob, "The Hyperledger fabric as a Blockchain framework preserves the security of electronic health records," *Frontiers in Public Health*, vol. 11, Art. no. 1272787, Nov. 2023. doi: 10.3389/fpubh.2023.1272787.

[43] O. Kuznetsov, P. Sernani, L. Romeo, E. Frontoni, and A. Mancini, "On the Integration of Artificial Intelligence and Blockchain Technology: A Perspective About Security," *IEEE Access*, vol. 12, pp. 3881–3897, 2024. doi: 10.1109/ACCESS.2023.3349019.

[44] Republic of South Africa, "Protection of Personal Information Act 4 of 2013 (POPIA)." Government Gazette, vol. 581, no. 37067, 26 Nov. 2013.

[45] M. de Bruyn, "The Protection of Personal Information (POPI) Act – Impact on South Africa," *International Business & Economics Research Journal*, vol. 13, no. 6, pp. 1315–1340, 2014.

[46] Republic of South Africa, "Commencement of certain sections of the Protection of Personal Information Act, 2013 (Act No. 4 of 2013)." Proclamation No. R. 21 of 2020, Government Gazette, vol. 660, no. 43461, 22 June 2020.

[47] M. Finck, *Blockchain and the General Data Protection Regulation: Can Distributed Ledgers Be Squared with European Data Protection Law?* Brussels, Belgium: European Parliamentary Research Service, Scientific Foresight Unit (STOA), PE 634.445, July 2019.

[48] B. Sibanda, B. Basheka, and J. van Romburgh, "Enhancing governance through blockchain technology in the South African public sector," *Africa's Public Service Delivery and Performance Review*, vol. 12, no. 1, Art. no. a734, Jan. 2024. doi: 10.4102/apsdpr.v12i1.734.

[49] M. Farhan, U. Butt, R. B. Sulaiman, and M. Alraja, "Self-sovereign identities and content provenance: VeriTrust—A blockchain-based framework for fake news detection," *Future Internet*, vol. 17, no. 10, Art. no. 448, 2025. doi: 10.3390/fi17100448.

[50] T. Zhu *et al.*, "Research on an On-Chain and Off-Chain Collaborative Storage Method Based on Blockchain and IPFS," *Future Internet*, vol. 18, no. 2, Art. no. 92, Feb. 2026. doi: 10.3390/fi18020092.

[51] A. de Jager, V. Marivate, and A. Modupe, "Multimodal misinformation detection in a South African social media environment," in *Artificial Intelligence Research (SACAIR 2023)*, Springer, 2023. arXiv:2312.04052.

[52] J. Zhang, Y. Huang, S. Liu, Y. Gao, and X. Hu, "Do BERT-Like Bidirectional Models Still Perform Better on Text Classification in the Era of LLMs?," in *Findings of the Association for Computational Linguistics: EMNLP 2025*, (Suzhou, China), pp. 18980–18989, Nov. 2025. doi: 10.18653/v1/2025.findings-emnlp.1033.

[53] M. Nadăş, L. Dioşan, and A. Tomescu, "Synthetic Data Generation Using Large Language Models: Advances in Text and Code," *IEEE Access*, vol. 13, pp. 134615–134633, 2025. doi: 10.1109/ACCESS.2025.3589503.

[54] R. Shuttleworth, J. Andreas, A. Torralba, and P. Sharma, "LoRA vs Full Fine-tuning: An Illusion of Equivalence," in *Advances in Neural Information Processing Systems 38 (NeurIPS 2025)*, pp. 193421–193456, 2025. doi: 10.52202/085713-5808.

[55] D. Tith, J.-S. Lee, H. Suzuki, W. M. A. B. Wijesundara, N. Taira, T. Obi, and N. Ohyama, "Application of blockchain to maintaining patient records in electronic health record for enhanced privacy, scalability, and availability," *Healthcare Informatics Research*, vol. 26, no. 1, pp. 3–12, 2020. doi: 10.4258/hir.2020.26.1.3.

[56] P. Ramazhamba and H. Venter, "A blockchain model for sharing information in criminal justice systems," in *Advances in Digital Forensics XIX* (G. Peterson and S. Shenoi, eds.), pp. 289–309, Cham, Switzerland: Springer, 2023.

[57] R. Amin, R. A. Chowdhury, S. M. Tanjim, A. Islam, and M. S. Islam, "xCRM: Blockchain interoperable crime report management system by utilizing Hyperledger Cacti & private data collection (PDC)," in *Proc. 2023 International Conference on Next-Generation Computing, IoT and Machine Learning (NCIM)*, 2023. doi: 10.1109/NCIM59001.2023.10212677.

[58] E. M. Alotaibi, H. Issa, and M. Codesso, "Blockchain-based conceptual model for enhanced transparency in government records: A

design science research approach," *International Journal of Information Management Data Insights*, vol. 5, Art. no. 100304, 2025. doi: 10.1016/j.jjimei.2024.100304.

[59] P. Mishra and R. Ganesan, "Securing e-governance: A blockchain-based framework for tamper-proof PDF document exchange," *Frontiers in Blockchain*, vol. 8, Art. no. 1699773, 2025. doi: 10.3389/fbloc.2025.1699773.

[60] H. C. Rashmi and S. Harish, "A secure and transparent model for academic credential management using blockchain technology," *International Journal of Information Technology and Management Information Systems*, vol. 17, no. 1, pp. 17–35, 2026. doi: 10.34218/IJITMIS_17_01_002.

[61] Y. Dai, G. Lu, and Y. Huang, "A blockchain-based access control system for secure and efficient hazardous material supply chains," *Mathematics*, vol. 12, no. 17, Art. no. 2702, 2024. doi: 10.3390/math12172702.

[62] M. A. Tasnim, A. Al Omar, M. S. Rahman, and M. Z. A. Bhuiyan, "CRAB: Blockchain based criminal record management system," in *Security, Privacy, and Anonymity in Computation, Communication, and Storage (SpaCCS 2018)* (G. Wang, J. Chen, and L. T. Yang, eds.), vol. 11342 of *Lecture Notes in Computer Science*, pp. 294–303, Cham, Switzerland: Springer, 2018. doi: 10.1007/978-3-030-05345-1_25.

[63] Google News Initiative, "Fact check images on Fact Check Explorer." Google News Initiative Training Center. Accessed: Oct. 2, 2026. [Online]. Available: https://newsinitiative.withgoogle.com/resources/trainings/fact-check-explorer/.

[64] International Fact-Checking Network, "The CoronaVirusFacts/DatosCoronaVirus Alliance Database." Poynter Institute. Accessed: Oct. 2, 2026. [Online]. Available: https://www.poynter.org/ifcn-covid-19-misinformation/.

[65] K. Peffers, T. Tuunanen, M. A. Rothenberger, and S. Chatterjee, "A design science research methodology for information systems research," *Journal of Management Information Systems*, vol. 24, no. 3, pp. 45–77, 2007.

[66] A. R. Hevner, S. T. March, J. Park, and S. Ram, "Design science in information systems research," *MIS Quarterly*, vol. 28, no. 1, pp. 75–105, 2004.

[67] A. Lombard, S. Mawere, T. Aina, E. Wolff, S. Gumede, E. Novick, F. Meyer, and J. Buys, "MzansiText and MzansiLM: An Open Corpus and Decoder-Only Language Model for South African Languages," in *Proceedings of the Fifteenth Language Resources and Evaluation Conference (LREC 2026)*, pp. 1394–1408, May 2026. doi: 10.63317/4g7i3kqszquh.

[68] J. O. Alabi, D. I. Adelani, M. Mosbach, and D. Klakow, "Adapting Pre-trained Language Models to African Languages via Multilingual Adaptive Fine-Tuning," in *Proceedings of the 29th International Conference on Computational Linguistics (COLING 2022)*, (Gyeongju, Republic of Korea), pp. 4336–4349, Oct. 2022.

[69] T. D. Belay *et al.*, "AfroXLMR-Social: Adapting Pre-trained Language Models for African Languages Social Media Text," in *Findings of the Association for Computational Linguistics: EMNLP 2025*, (Suzhou, China), pp. 15570–15587, Nov. 2025. doi: 10.18653/v1/2025.findings-emnlp.842.

[70] E. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, and W. Chen, "LoRA: Low-Rank Adaptation of Large Language Models," in *Proceedings of the 10th International Conference on Learning Representations (ICLR 2022)*, 2022.

[71] M. Szep, D. Rueckert, R. von Eisenhart-Rothe, and F. Hinterwimmer, "Fine-tuning Large Language Models with Limited Data: A Survey and Practical Guide," *Transactions of the Association for Computational Linguistics*, vol. 14, pp. 341–377, 2026. doi: 10.1162/tacl.a.627.

[72] A. Tiwari, O. Ogunremu, T. Oko-odion, J. Egbewale, and H. S. Nwokocha, "Sample-Size Scaling of the African Languages NLI Evaluation," in *Proceedings of the 7th Workshop on African Natural Language Processing (AfricaNLP 2026)*, pp. 217–227, Mar. 2026. doi: 10.18653/v1/2026.africanlp-main.22.

[73] G. Wang *et al.*, "Consistency and Completeness: Rethinking Distributed Stream Processing in Apache Kafka," in *Proc. 2021 Int. Conf. Management of Data (SIGMOD '21)*, pp. 2602–2613, 2021. doi: 10.1145/3448016.3457556.

[74] A. Akanbi, "ESTemd: A Distributed Processing Framework for Environmental Monitoring based on Apache Kafka Streaming Engine." arXiv:2104.01082, 2021.

[75] Apache Software Foundation, "Introduction to Apache Kafka." Apache Kafka Documentation, 2026. Accessed: Sep. 15, 2026. [Online]. Available: https://apache.org.

[76] E. Androulaki *et al.*, "Hyperledger Fabric: A distributed operating system for permissioned blockchains," in *Proc. 13th EuroSys Conf. (EuroSys '18)*, pp. 1–15, 2018.

[77] S. Nakamoto, "Bitcoin: A Peer-to-Peer Electronic Cash System," 2008. [Online]. Available: https://bitcoin.org/bitcoin.pdf.

[78] V. Buterin, "A Next-Generation Smart Contract and Decentralized Application Platform." Ethereum White Paper, 2014. [Online]. Available: https://ethereum.org/en/whitepaper/.

[79] B. Meister and H. Price, "Gas fees on the Ethereum blockchain: from foundations to derivative valuations," *Frontiers in Blockchain*, vol. 7, Art. no. 1462666, 2024. doi: 10.3389/fbloc.2024.1462666.

[80] S. Dhall, A. D. Dwivedi, S. K. Pal, and G. Srivastava, "Blockchain-based framework for reducing fake or vicious news spread on social media/messaging platforms," *ACM Transactions on Asian and Low-Resource Language Information Processing*, vol. 21, no. 1, pp. 1–33, 2021. doi: 10.1145/3467019.

[81] H. Han, R. K. Shiwakoti, R. Jarvis, C. Mordi, and D. Botchie, "Accounting and auditing with blockchain technology and artificial intelligence: A literature review," *International Journal of Accounting Information Systems*, vol. 48, Art. no. 100598, 2023. doi: 10.1016/j.accinf.2022.100598.

[82] A. Shcherbakov, "Hyperledger Indy Besu as a permissioned ledger in self-sovereign identity," in *Open Identity Summit 2024*, Lecture Notes in Informatics, pp. 127–137, Bonn, Germany: Gesellschaft für Informatik e.V., 2024. doi: 10.18420/OID2024_11.

[83] J. Polge, J. Robert, and Y. Le Traon, "Permissioned blockchain frameworks in the industry: A comparison," *ICT Express*, vol. 7, 2021. doi: 10.1016/j.icte.2020.09.002.

[84] A. Pericherla, P. Paul, S. Sural, J. Vaidya, and V. Atluri, "Towards Supporting Attribute-Based Access Control in Hyperledger Fabric Blockchain," in *IFIP Advances in Information and Communication Technology*, vol. 648, pp. 360–376, Cham, Switzerland: Springer, 2022. doi: 10.1007/978-3-031-06975-8_21.

[85] R3, "Trade-offs - Enterprise 4.14." R3 Documentation, 2026. [Online]. Available: https://docs.r3.com/en/platform/corda/4.14/enterprise/key-concepts-tradeoffs.html.

[86] ConsenSys, "Enhanced Permissions Model." GoQuorum Documentation, 2023. [Online]. Available: https://goquorum.readthedocs.io/Permissioning/Enhanced%20Permissions%20Model/Overview/.

[87] MultiChain, "MultiChain permissions management," 2026. [Online]. Available: https://www.multichain.com/developers/permissions-management/.

[88] J. Benet, "IPFS - Content Addressed, Versioned, P2P File System." arXiv:1407.3561, 2014.

[89] M. Avelino and A. A. d. A. Rocha, "BlockProof: A Framework for Verifying Authenticity and Integrity of Web Content," *Sensors*, vol. 22, no. 3, Art. no. 1165, 2022. doi: 10.3390/s22031165.

[90] D. Ongaro and J. Ousterhout, "In search of an understandable consensus algorithm," in *Proc. 2014 USENIX Annual Technical Conference (USENIX ATC '14)*, pp. 305–320, 2014.

[91] Hyperledger Fabric, "The Ordering Service." Hyperledger Fabric Documentation, 2024. [Online]. Available: https://hyperledger-fabric.readthedocs.io/en/latest/orderer/ordering_service.html.

[92] U. Kilic and M. K. Keles, "Performance And Scalability Comparison of Raft And SmartBFT Consensus Mechanisms In A Private Blockchain Network," *Control Engineering and Applied Informatics*, vol. 28, no. 2, 2026.

[93] P. Ndayizigamiye and S. Dube, "Performance Evaluation of Hyperledger Fabric Blockchain Network with PBFT and Raft Consensus Protocols," *East African Journal of Information Technology*, vol. 7, no. 1, 2024. [Online]. Available: https://journals.eanso.org/index.php/eajit/article/view/3970.

[94] P. Thakkar, S. Nathan, and B. Viswanathan, "Performance Benchmarking and Optimizing Hyperledger Fabric Blockchain Platform," in *Proc. 2018 IEEE 26th Int. Symp. Modeling, Analysis, and Simulation of Computer and Telecommunication Systems (MASCOTS)*, (Milwaukee, WI, USA), pp. 264–276, 2018. doi: 10.1109/MASCOTS.2018.00034.

[95] A. A. Salih, *Byzantine fault tolerant consensus for hyperledger fabric*. Ph.d. dissertation, University of North Carolina at Charlotte, Charlotte, NC, USA, 2024.

[96] R. Kaushal and N. Kumar, "Exploring Hyperledger Caliper Benchmarking Tool to Measure the Performance of Blockchain Based Solutions," in *Proc. 2024 Int. Conf. Reliability, Infocom Technologies and Optimization (ICRITO)*, pp. 1–6, 2024. doi: 10.1109/ICRITO61523.2024.10522188.

[97] F. C. Geyer, H.-A. Jacobsen, R. Mayer, and P. Mandl, "An End-to-End Performance Comparison of Seven Permissioned Blockchain Systems," in *Proc. 24th Int. Middleware Conf. (Middleware '23)*, pp. 71–84, 2023. doi: 10.1145/3590140.3629106.